

Supporting Information

Incorporating Synthetic Accessibility in Drug Design: Predicting Reaction Yields of Suzuki Cross-Couplings by Leveraging AbbVie's 15-Year Parallel Library Data Set

Priyanka Raghavan^{‡a}, Alexander J. Rago^{‡b}, Pritha Verma^b, Majdi M. Hassan^c, Gashaw M. Goshu^b, Amanda W. Dombrowski^b, Abhishek Pandey^c, Connor W. Coley^{*a}, and Ying Wang^{*b}

^aDepartment of Chemical Engineering, Massachusetts Institute of Technology, 77 Massachusetts Ave, Cambridge, MA 02139, United States; ^bAdvanced Chemistry Technologies Group, AbbVie, Inc., 1 N Waukegan Rd, North Chicago, IL 60064, United States; ^cRAIDERS Group, AbbVie, Inc., 1 N Waukegan Rd, North Chicago, IL 60064, United States

[‡]P.R. and A.J.R. contributed equally to this work.

*Email: ccoley@mit.edu; wang.ying@abbvie.com

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1. General Information

General Experimental Considerations:

All Suzuki cross-coupling reactions were sealed under air atmosphere under standard benchtop conditions. Dimethyl sulfoxide (472301-4L, Millipore Sigma) was purchased and used to prepare a 9:1 DMSO/water solution for UPLC-MS plate dilutions and preparative HPLC sample preparation. Pd(dppf)Cl₂·CH₂Cl₂ (93-2801, Sigma-Aldrich), Cs₂CO₃ (050215, Strem), and dioxane (296309-100ML, Sigma-Aldrich) were used as received. Siliacat DPP-Pd (R390-100, SiliCycle) was stored in a 5 °C laboratory refrigerator and was otherwise used as received. Water used in the reactions was deionized using a Barnstead NANOpure Diamond water purifier. The aryl halides and boronic acid/esters were sourced either commercially or from AbbVie's internal chemical repository and directly used as received (please see **Figure S21** and **Figure S22** for specific sources). For analysis of the crude reactions, filtration was conducted using a 350 µL GHP 96-well filter plate (Pall Laboratories) into an LCMS plate (200 µL volume, V-bottom or U-bottom 96-well plate, Corning, part number 29444-102). All centrifugation steps were conducted using a ThermoScientific Multifuge X Pro. A Biotage Initiator⁺ with a 0.5-2.0 mL microwave vial was used to conduct the four reactions that required microwave irradiation; otherwise, an IKA Plate (RCT digital) Magnetic Stirrer (with a metal 4mL vial heating block) was used to heat all other reactions.

Crude and final product purity checks were performed on a Waters Acquity UPLC system equipped with an in-line photodiode array detector (PDA) and an SQD mass spectrometer, running MassLynx 4.1 and Openlynx 4.1 software (Waters Corporation, Milford, MA, USA). The SQD mass spectrometer was operated under positive ESI ionization conditions. The column used was a Waters BEH C8, 1.7 µm (2.1 mm × 30 mm) at a temperature of 55 °C. The following ammonium acetate analytical method was used: a gradient of 10-100% acetonitrile (A) and 10 mM ammonium acetate in water (B) was used, at a flow rate of 1.0 mL/min (0-0.1 min 10% A, 0.1-1.1 min 10- S3 100% A, 1.1-1.3 min 100% A, 1.3-1.4 min 100-10% A; for FA methods, B was substituted with 0.1 v/v% FA in water).

Isolation of products from reaction mixtures was accomplished with a preparative-scale reverse-phase HPLC instrument. When possible, an ammonium acetate purification method was used: for example, a gradient of acetonitrile (A) and 10 mM ammonium acetate in water (B) was used, at a flow rate of 50 mL/min (0-0.5 min 5% A, 0.5-8.5 min linear gradient 05-100% A, 8.7-10.7 min 100% A, 10.7-11 min linear gradient 100- 05% A). A formic acid method was used for samples which required repurification: for example, a gradient of acetonitrile (A) and 0.1% formic acid in water (B) was used, at a flow rate of 50 mL/min (0-0.5 min 5% A, 0.5-8.5 min linear gradient 05-100% A, 8.7-10.7 min 100% A, 10.7-11 min linear gradient 100-05% A). Focused gradients were also employed for more challenging separations. Samples were injected in 1.5-2.0 mL DMSO. The main instrument used was a Waters Prep-HPLC Purification system (small scale (10mg-300mg)): Samples were purified by preparative HPLC on a Phenomenex Luna C8(2) 5 µm 100Å AXIA column (three 30 mm × 75 mm columns connected in sequence). General gradients follow. "Formic acid method": A gradient of acetonitrile (A) and 0.1% formic acid in water (B) was used, at a flow rate of 50 mL/min (0-1.0 min 5% A, 1.0-10.5 min linear gradient 5-100% A, 10.5-14.0min 100% A). "Ammonium acetate method": A gradient of acetonitrile (A) and 10 mM ammonium acetate in water (B) was used, at a flow rate of 50mL/min (0-1.0 min 5% A, 1.0-10.5 min linear gradient 5-100% A, 10.5-14.0 min 100% A). For one re-purification sample, an Agilent 1100 Series Purification system was used, consisting of 2-coupled C8 5 µm 100 Å columns (30 mm × 75 mm each) and the following modules: Agilent 1100 Series LC/MSD SL mass spectrometer with API-electrospray source; two Agilent 1100 Series preparative pumps; Agilent 1100 Series isocratic pump; Agilent 1100 Series diode array detector with preparative (0.3 mm) flow cell; Agilent active-splitter, IFC-PAL fraction collector/autosampler. The make-up pump for the mass spectrometer used 3/1 methanol:water with 0.1%

formic acid at a flow rate of 1 mL/min. Fraction collection was automatically triggered when the extracted ion chromatogram (EIC) for the target mass exceeded the threshold specified in the method. The system was controlled using Agilent Chemstation (Rev B.10.03), Agilent A2Prep, and Leap FractPal software, with custom Chemstation macros for data export. For one re-purification, a custom purification system was used which consisted of two Kinetix 5 μ m C18 100 Å columns (50mm x 30 mm) connected in series with the following modules: Gilson 305 and 306 pumps; Gilson 806 Manometric module; Gilson UV/Vis 155 detector; Gilson 506C interface box; Gilson FC204 fraction collector; Agilent G1968D Active Splitter; Thermo MSQ Plus mass spectrometer. A flow rate of 40 mL/min was used. The system was controlled through a combination of Thermo Xcalibur 2.0.7 software and a custom application written in-house using Microsoft Visual Basic 6.0.

NMR spectra were acquired on two Bruker Avance III HD 500 MHz spectrometers with DCH cryoprobes, a 600 MHz spectrometer, or a 400 MHz spectrometer, all equipped with sample changers. Samples were also acquired on two 400 MHz Varian MR-400 spectrometers, particularly those conducted at 90 °C. Chemical shifts are reported as parts per million (ppm) downfield from tetramethylsilane and referenced to the solvent (e.g. DMSO-*d*₆, pyridine-*d*₅, CDCl₃). NMR data are represented as follows: chemical shift, multiplicity (s = singlet, br s = broad singlet, d = doublet, dd = doublet of doublet, t = triplet, q = quartet, p = quintet, h = sextet, hept = septet, dt = doublet of triplet of triplets, qd = quartet of doublets, qt = quartet of triplets, dtd = doublet of doublet of triplet of doublets, m = multiplet, etc.), coupling constant, and integration. Coupling constants (J) are given in Hertz (Hz). The temperature was recorded for samples conducted at elevated temperatures, otherwise the samples were acquired at room temperature. Low-resolution mass spectrometry (LRMS) characterization was collected during the final purity check with the Waters Acquity UPLC system described previously with positive ESI ionization.

Machine learning was conducted using a workstation with the following configuration: 128 threads, 64 Core CPU (AMD Ryzen Threadripper PRO 3995WX 64-Cores); 3 x GPUs (NVIDIA GeForce RTX 3090); 512GB DDR4 RAM; 1x1TB SSD; 1 x 2TB SSD.

2. Machine Learning Experimental Details

Disclaimer: This code was tested on a few different machines, and we observed slight numerical variations across different hardware architectures. None of these variations were found to change the conclusions reached in this work.

Code Availability: Code is available at https://github.com/priyanka-rag/suzuki_yield_predict_external.

a. Dataset availability

The dataset used in this study was assembled from library reactions conducted over a >15-year period in support of various AbbVie medicinal chemistry programs. As such, the structures present in the dataset are proprietary and cannot be disclosed. Therefore, the one-hot encoded version of the dataset is available as an additional file. The full structures of the prospective monomer replacement application are shown.

b. Dataset processing

The dataset used in this study was obtained by processing CSV files retrieved from AbbVie's library information management system (LIMS) database using Python code written in Jupyter notebook files. The final dataset includes the columns: Library, Core_Smiles, Monomer_Smiles, Product_Smiles, Percent_Yield, Catalyst, Base, Ligand, Solvent, Solvent_Unsplit (not used for modeling), Aryl_Halide,

Boronic_Acid. As stated previously, the SMILES have been replaced with OHE representations for the version of the dataset that has been released.

c. Model details and hyperparameter optimization

Base model details: The Random Forest (RF) model was provided by the Sci-kit learn package, and the Gradient Boosting (xGB) model was provided by Kaggle. For both, default models were used with the exception of a few tuned hyperparameters as described in the next section. The feed-forward neural network model (NN) was implemented using the pytorch package. The base architecture consists of 3 fully-connected hidden layers, and model training was conducted for a maximum of 200 epochs using an Adam optimizer and learning rate scheduler with a patience of 3 epochs and decrease factor of 0.5. Early stopping was implemented with a patience of 10 epochs.

Reaction modeling details: In all cases, concatenated features of the boronic acid and aryl halide for each reaction were used. Additionally, OHE features for the catalyst + ligand system and base, and MHE (multi-hot encoding) features of the solvent were concatenated to the molecular features. Note that OHE/MHE representations of reaction conditions were used even for fingerprint or DFT substrate representations.

Reaction splitting details: For the retrospective analysis, 15 dataset splits (each 85% trainval/15% test) were created for each split type. For each, the trainval set was then additionally split into 70% train/15% val for the neural network training.

Hyperparameter optimization: An exhaustive hyperparameter optimization campaign was conducted for each combination of model, featurization, task, and split type. This led to a total of $3 \times 4 \times 3 \times 3 = 108$ hyperparameter optimizations. For each, the 85% trainval set was further split into 5-70% train/15% val sets, the model was trained and tested on each of the 5 sets, and the average metric over all 5 sets was optimized for. The binary and multi-class classification tasks used accuracy as the metric, and the regression task used the R^2 value. Each optimization was conducted on the first 85% trainval/15% test dataset split of each type, and once the optimal parameters were found, a model using these parameters was trained on the entire trainval set and tested on the corresponding test set for all 15 dataset splits to get the final reported values, as shown in **Figure 5** of the main text. Note that the core and monomer-based splits will not maintain this precise 70%/15%/15% ratio, due to the added splitting constraint of making sure certain structures are not repeated across the train/val/test sets. Still, we have tried to make the splitting ratio close to these values for all splits.

All optimizations were conducted over 25 trials using the *hyperopt* package in Python. The chosen parameter space for each model, compared to default hyperparameters, is shown below:

	<u>Choices/Range</u>	<u>Default</u>
<u>Random Forest:</u>		
· <u>n_estimators</u>	[20,500]	100
<u>Gradient Boosting:</u>		
· <u>n_estimators</u>	[20,500]	100
· <u>learning_rate</u>	[1e-4, 1e-1]	0.1
<u>Neural Network:</u>		
· <u>Architecture</u>	3 layers, ea. 64-1024 bits	N/A
· <u>init_learning_rate</u>	[1e-5, 1e-2]	1e-3

Table S1. Hyperparameters and Scan Ranges Used for Optimization

The 108 sets of selected hyperparameters can be found on our Github page under the following directory: `dataset_files/retrospective/optimal_hyperparameters.csv`.

d. DFT pipeline and descriptor histograms

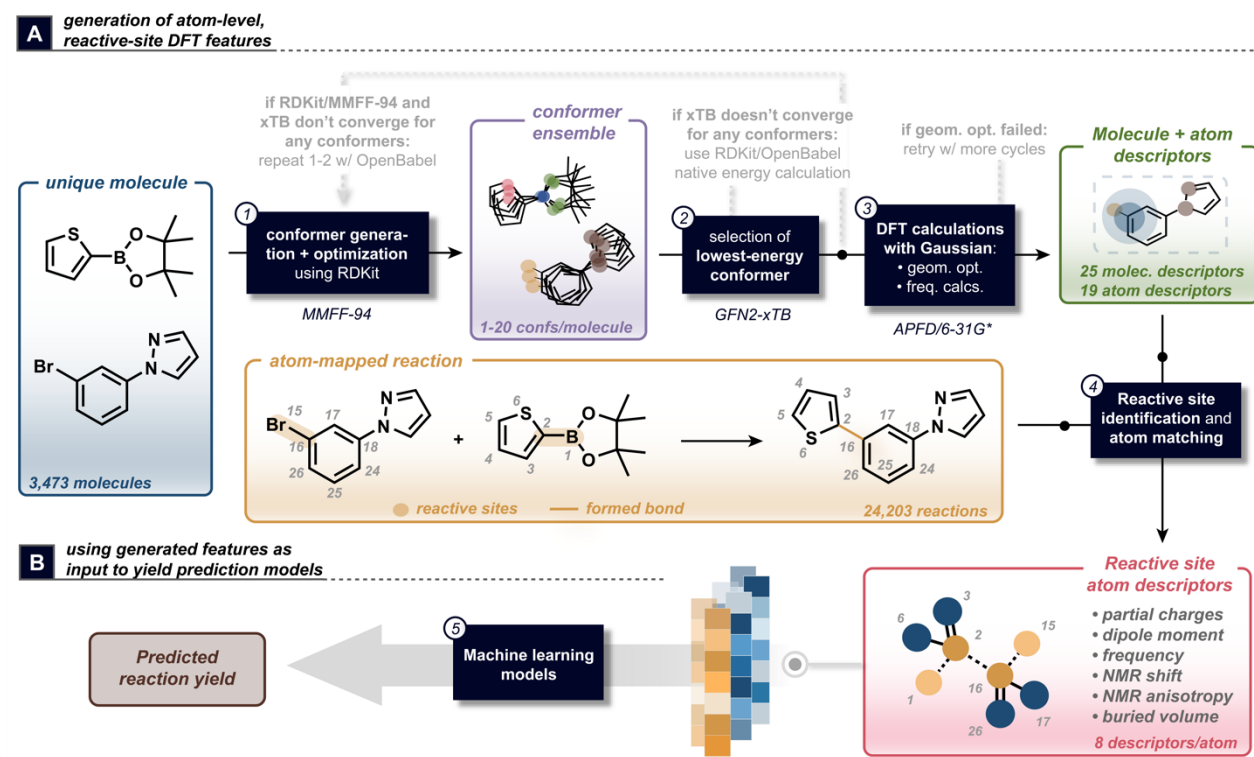


Figure S1. Full DFT descriptor generation pipeline, consisting of a multi-stage process of conformer generation, lowest-energy conformer selection, descriptor generation using Gaussian 16, and extraction of molecule-level and atom-level, reactive-site DFT features. Error handling and modes are shown in gray.

The full process to generate the DFT descriptors that were used as input to the ML models is detailed as follows:

1. All reactions in the dataset were atom-mapped, and the unique aryl halide and boronic acid reactants were extracted.
2. For each reactant, conformers were generated in RDKit¹ [<https://www.rdkit.org>] and initial optimization was conducted using molecular force field methods (*MMFF-94*). Conformers that did not converge during the *MMFF-94* optimization were pruned at this stage. This resulted in 1-20 conformers per molecule. RDKit parameters used for conformer generation are given below (note: these are the same parameters used in Auto-QChem²):
 - useSymmetryForPruning = True
 - useSmallRingTorsions = True
 - useMacrocycleTorsions = True
 - ETversion = 2 (default = 1)
 - pruneRmsThresh = 0.35 (default = -1.0)
 - Max. # of conformers = 20 (default = 10)

3. Conformer energy was then calculated semi-empirically using xTB at the GFN2 level of theory.³ Conformers for which the xTB energy calculation was unsuccessful were pruned at this stage. If this energy calculation was unsuccessful for all conformers for a molecule, the calculation was repeated using RDKit's native method *calcEnergy()*. If no valid conformers remained after *MMFF-94* optimization and xTB calculation, the pipeline was repeated from Step 2, but using Open Babel *gen3D* conformer generation⁴ rather than RDKit. Subsequently, the lowest-energy valid conformer for each molecule was then selected.
4. Gaussian16⁵ geometry optimization and frequency calculations were then used to calculate several DFT descriptors for each lowest-energy conformer. The set of Gaussian input commands used for each molecule is given below:
 - # opt=CalcFc APFD/6-31G* scf=xqc
 - freq APFD/6-31G* volume NMR pop=NPA density=current Geom=AllCheck Guess=Read
 - TD(NStates=10, Root=1) APFD/6-31G* volume pop=NPA density=current Geom=AllCheck Guess=Read

Most conformers were successfully run through Gaussian the first time around. Among those that failed, most were able to be corrected through increasing the maximum number of cycles allotted for geometry optimization. Reactions involving the final molecules that failed were removed from the dataset.

5. 25 molecular descriptors per molecule and 19 atom descriptors per atom per molecule were extracted. Then, for each atom-mapped reaction, the 4 atoms of the reaction site (aryl halide halogen and connected carbon, boronic acid boron and connected carbon) were identified and matched to the corresponding extracted atom-level descriptors. Example histograms of some pertinent molecule and atom-level DFT descriptors across all reactions are shown below:

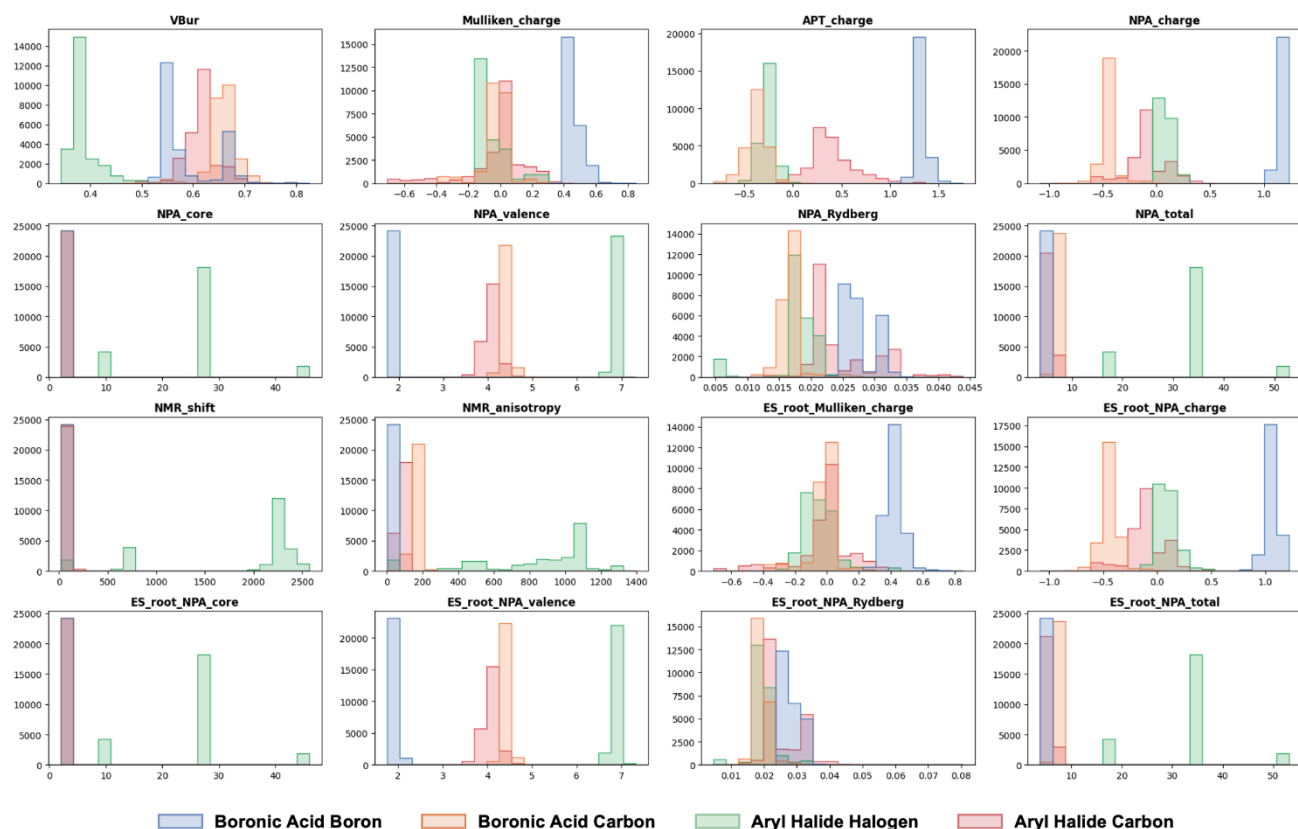


Figure S2. Histograms of 16 atom-level descriptors across all reactions, colored by each type of atom in the reactive site.

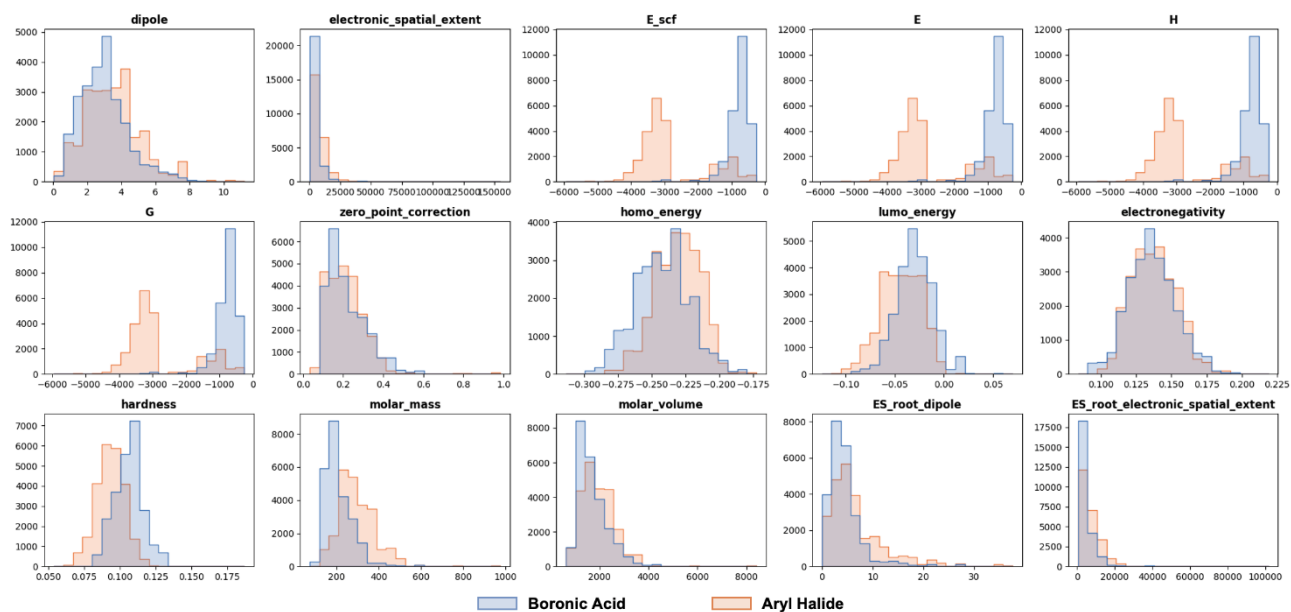


Figure S3. Histograms of 15 molecule-level descriptors across all reactions, colored by each type of molecule (boronic acid or aryl halide).

6. Descriptors were mostly chosen based on those that exhibited varied distributions across the dataset. The atom-level descriptors were concatenated with molecule-level descriptors to form the final feature vector of length 48 for each reaction, used as input to the models. The exact atom and molecule-level descriptors used for modeling are shown below:
- Atom descriptors: buried volume (VBur), Mulliken charge (Mulliken_charge), atomic polar tensor charge (APT_charge), natural population analysis charge (NPA_charge), natural population analysis valence (NPA_valence), NMR anisotropy (NMR_anisotropy), ES_root_Mulliken_charge, ES_root_NPA_charge
 - Molecule descriptors: dipole (dipole), molar mass (molar_mass), self-consistent field energy (E_scf), electronic spatial extent (electronic_spatial_extent), HOMO energy (homo_energy), LUMO energy (lumo_energy), electronegativity, Gibbs free energy (G)

e. Full performance tables and additional metrics

Note that the MAE/RMSE in the following tables were calculated using yield values divided by 100 (i.e. scaled between 0 and 1).

Split	Model	Binary Classification					Multi-Class	Regression		
		Accuracy	AUROC	AUPRC	Precision	Recall	Accuracy	R ²	MAE	RMSE
Random	RF	0.739 ± 0.006	0.747 ± 0.008	0.837 ± 0.008	0.761 ± 0.005	0.899 ± 0.006	0.494 ± 0.007	-0.015 ± 0.029	0.120 ± 0.002	0.178 ± 0.003
	xGB	0.737 ± 0.006	0.755 ± 0.004	0.853 ± 0.003	0.739 ± 0.006	0.949 ± 0.005	0.501 ± 0.007	0.284 ± 0.011	0.116 ± 0.002	0.149 ± 0.002
	NN	0.743 ± 0.006	0.769 ± 0.004	0.866 ± 0.004	0.786 ± 0.005	0.856 ± 0.009	0.522 ± 0.004	0.340 ± 0.011	0.106 ± 0.002	0.143 ± 0.002
Core	RF	0.711 ± 0.020	0.672 ± 0.016	0.772 ± 0.024	0.719 ± 0.024	0.932 ± 0.010	0.384 ± 0.015	-0.183 ± 0.066	0.132 ± 0.009	0.190 ± 0.013
	xGB	0.711 ± 0.020	0.669 ± 0.021	0.783 ± 0.027	0.707 ± 0.025	0.970 ± 0.006	0.375 ± 0.019	0.078 ± 0.029	0.132 ± 0.007	0.167 ± 0.010
	NN	0.708 ± 0.022	0.688 ± 0.032	0.794 ± 0.030	0.728 ± 0.033	0.904 ± 0.043	0.423 ± 0.016	0.090 ± 0.038	0.128 ± 0.008	0.166 ± 0.009
Monomer	RF	0.698 ± 0.020	0.657 ± 0.007	0.775 ± 0.019	0.712 ± 0.024	0.929 ± 0.008	0.460 ± 0.011	-0.187 ± 0.039	0.132 ± 0.004	0.193 ± 0.005
	xGB	0.694 ± 0.022	0.652 ± 0.010	0.777 ± 0.024	0.696 ± 0.024	0.973 ± 0.004	0.448 ± 0.010	0.213 ± 0.016	0.126 ± 0.002	0.157 ± 0.003
	NN	0.685 ± 0.026	0.659 ± 0.022	0.785 ± 0.033	0.712 ± 0.043	0.909 ± 0.062	0.453 ± 0.008	0.207 ± 0.018	0.121 ± 0.002	0.158 ± 0.003

Table S2. Performance across 15 dataset splits using one-hot encoding (OHE) features.

Split	Model	Binary Classification					Multi-Class	Regression		
		Accuracy	AUROC	AUPRC	Precision	Recall	Accuracy	R ²	MAE	RMSE
Random	RF	0.768 ± 0.006	0.801 ± 0.005	0.882 ± 0.004	0.789 ± 0.005	0.900 ± 0.005	0.537 ± 0.006	0.357 ± 0.016	0.102 ± 0.001	0.141 ± 0.002
	xGB	0.768 ± 0.006	0.799 ± 0.006	0.879 ± 0.004	0.782 ± 0.005	0.915 ± 0.004	0.546 ± 0.006	0.377 ± 0.012	0.105 ± 0.001	0.139 ± 0.002
	NN	0.764 ± 0.005	0.793 ± 0.004	0.871 ± 0.005	0.804 ± 0.005	0.863 ± 0.010	0.544 ± 0.005	0.382 ± 0.013	0.102 ± 0.002	0.139 ± 0.002
Core	RF	0.724 ± 0.018	0.712 ± 0.018	0.807 ± 0.029	0.732 ± 0.023	0.925 ± 0.010	0.435 ± 0.015	0.097 ± 0.079	0.127 ± 0.006	0.165 ± 0.009
	xGB	0.719 ± 0.018	0.702 ± 0.021	0.801 ± 0.027	0.732 ± 0.024	0.915 ± 0.019	0.428 ± 0.019	0.081 ± 0.053	0.131 ± 0.008	0.167 ± 0.010
	NN	0.714 ± 0.016	0.689 ± 0.017	0.791 ± 0.027	0.733 ± 0.024	0.899 ± 0.017	0.421 ± 0.023	0.097 ± 0.063	0.127 ± 0.007	0.166 ± 0.008
Monomer	RF	0.745 ± 0.012	0.769 ± 0.012	0.859 ± 0.015	0.761 ± 0.020	0.909 ± 0.013	0.520 ± 0.012	0.312 ± 0.024	0.107 ± 0.003	0.147 ± 0.003
	xGB	0.742 ± 0.014	0.758 ± 0.014	0.851 ± 0.019	0.762 ± 0.021	0.901 ± 0.013	0.517 ± 0.013	0.315 ± 0.027	0.111 ± 0.003	0.146 ± 0.003
	NN	0.737 ± 0.013	0.753 ± 0.016	0.842 ± 0.017	0.782 ± 0.019	0.848 ± 0.023	0.513 ± 0.013	0.324 ± 0.024	0.108 ± 0.002	0.145 ± 0.003

Table S3. Performance across 15 dataset splits using fingerprint (fp) features.

Split	Model	Binary Classification					Multi-Class	Regression		
		Accuracy	AUROC	AUPRC	Precision	Recall	Accuracy	R ²	MAE	RMSE
Random	RF	0.766 ± 0.005	0.794 ± 0.006	0.876 ± 0.005	0.787 ± 0.005	0.899 ± 0.006	0.542 ± 0.007	0.379 ± 0.013	0.103 ± 0.002	0.139 ± 0.002
	xGB	0.769 ± 0.004	0.797 ± 0.004	0.877 ± 0.004	0.793 ± 0.005	0.893 ± 0.006	0.545 ± 0.007	0.388 ± 0.011	0.102 ± 0.001	0.138 ± 0.002
	NN	0.739 ± 0.007	0.748 ± 0.008	0.841 ± 0.006	0.773 ± 0.007	0.874 ± 0.012	0.511 ± 0.004	0.311 ± 0.012	0.109 ± 0.001	0.146 ± 0.002
Core	RF	0.716 ± 0.016	0.713 ± 0.017	0.812 ± 0.027	0.736 ± 0.024	0.896 ± 0.023	0.431 ± 0.020	0.085 ± 0.058	0.132 ± 0.007	0.167 ± 0.008
	xGB	0.716 ± 0.019	0.702 ± 0.019	0.803 ± 0.029	0.736 ± 0.025	0.897 ± 0.020	0.423 ± 0.016	0.084 ± 0.065	0.129 ± 0.006	0.167 ± 0.007
	NN	0.689 ± 0.021	0.644 ± 0.026	0.754 ± 0.035	0.716 ± 0.028	0.890 ± 0.036	0.393 ± 0.017	0.049 ± 0.039	0.137 ± 0.008	0.170 ± 0.009
Monomer	RF	0.746 ± 0.015	0.761 ± 0.016	0.850 ± 0.016	0.764 ± 0.020	0.903 ± 0.020	0.526 ± 0.014	0.330 ± 0.024	0.110 ± 0.002	0.145 ± 0.003
	xGB	0.745 ± 0.016	0.756 ± 0.013	0.845 ± 0.018	0.772 ± 0.020	0.885 ± 0.019	0.518 ± 0.012	0.324 ± 0.022	0.109 ± 0.002	0.145 ± 0.002
	NN	0.730 ± 0.017	0.735 ± 0.016	0.829 ± 0.020	0.763 ± 0.020	0.871 ± 0.027	0.497 ± 0.018	0.292 ± 0.028	0.113 ± 0.003	0.149 ± 0.003

Table S4. Performance across 15 dataset splits using DFT features.

Split	Model	Binary Classification					Multi-Class	Regression		
		Accuracy	AUROC	AUPRC	Precision	Recall	Accuracy	R ²	MAE	RMSE
Random	RF	0.773 ± 0.005	0.808 ± 0.006	0.886 ± 0.005	0.789 ± 0.005	0.909 ± 0.005	0.551 ± 0.005	0.390 ± 0.015	0.101 ± 0.001	0.138 ± 0.002
	xGB	0.772 ± 0.005	0.803 ± 0.005	0.882 ± 0.004	0.791 ± 0.005	0.904 ± 0.006	0.548 ± 0.006	0.389 ± 0.012	0.103 ± 0.001	0.138 ± 0.002
	NN	0.765 ± 0.004	0.795 ± 0.003	0.873 ± 0.006	0.804 ± 0.006	0.867 ± 0.010	0.544 ± 0.006	0.387 ± 0.014	0.100 ± 0.001	0.138 ± 0.002
Core	RF	0.723 ± 0.016	0.720 ± 0.019	0.816 ± 0.029	0.734 ± 0.023	0.920 ± 0.012	0.449 ± 0.017	0.137 ± 0.055	0.126 ± 0.006	0.162 ± 0.008
	xGB	0.718 ± 0.017	0.707 ± 0.014	0.805 ± 0.025	0.740 ± 0.026	0.890 ± 0.021	0.432 ± 0.022	0.083 ± 0.048	0.130 ± 0.007	0.167 ± 0.008
	NN	0.712 ± 0.016	0.687 ± 0.016	0.790 ± 0.027	0.732 ± 0.024	0.898 ± 0.021	0.422 ± 0.023	0.105 ± 0.059	0.127 ± 0.007	0.165 ± 0.009
Monomer	RF	0.754 ± 0.013	0.783 ± 0.013	0.865 ± 0.015	0.767 ± 0.020	0.914 ± 0.016	0.537 ± 0.013	0.347 ± 0.023	0.107 ± 0.002	0.143 ± 0.003
	xGB	0.751 ± 0.014	0.770 ± 0.015	0.857 ± 0.017	0.771 ± 0.020	0.900 ± 0.014	0.524 ± 0.013	0.335 ± 0.025	0.109 ± 0.003	0.144 ± 0.003
	NN	0.739 ± 0.014	0.757 ± 0.017	0.846 ± 0.019	0.777 ± 0.021	0.863 ± 0.014	0.514 ± 0.013	0.327 ± 0.026	0.107 ± 0.002	0.145 ± 0.003

Table S5. Performance across 15 dataset splits using concatenated fingerprint and DFT (fp + DFT) features.

We additionally explored 2 alternate, more complex model and feature types, due to their recent increasing prevalence for chemical predictivity tasks. Hyperparameter optimization was not conducted in both cases.

Graph neural network (GNN) with condensed graph of reaction (CGR) representation:⁶ We used the base model from the latest version of Chemprop, utilizing our atom-mapped reactions for the CGR representation. The batch size was changed to 512; all other hyperparameters were kept default.

Model	Split Type	Binary Classification			Multi-Class	Regression
		Accuracy	AUROC	AUPRC	Accuracy	R ²
GNN (CGR rep.)	Random	0.762 ± 0.006	0.789 ± 0.004	0.873 ± 0.004	0.537 ± 0.008	0.378 ± 0.014
	Core	0.712 ± 0.028	0.692 ± 0.028	0.795 ± 0.032	0.421 ± 0.020	0.100 ± 0.048
	Monomer	0.746 ± 0.015	0.765 ± 0.015	0.853 ± 0.019	0.520 ± 0.010	0.325 ± 0.021

Table S6. Performance across 15 dataset splits using a Chemprop GNN with condensed graph of reaction representation.

While performance was comparable, albeit slightly lower, to that of the RF model, and may have surpassed it with hyperparameter optimization, the amount of time taken to train the GNN renders this model much more computationally expensive. Therefore, the RF model was chosen for the prospective studies.

Language-model features: These features represent embeddings extracted from the encoder of a transformer model for reaction product prediction trained on the Pistachio dataset. We used these features summed over tokens as direct inputs to our base retrospective models (RF, xGB, and NN). Default hyperparameters were used in all cases; for the neural network, a 3-hidden layer architecture (512/256/64 bits) was used.

Split	Model	Binary Classification			Multi-Class	Regression
		Accuracy	AUROC	AUPRC	Accuracy	R ²
Random	RF	0.717 ± 0.006	0.713 ± 0.008	0.821 ± 0.007	0.475 ± 0.006	0.235 ± 0.012
	xGB	0.710 ± 0.005	0.701 ± 0.009	0.817 ± 0.009	0.472 ± 0.006	0.197 ± 0.012
	NN	0.704 ± 0.006	0.682 ± 0.007	0.801 ± 0.007	0.457 ± 0.008	0.218 ± 0.011
Core	RF	0.675 ± 0.021	0.619 ± 0.019	0.750 ± 0.027	0.382 ± 0.013	0.041 ± 0.036
	xGB	0.662 ± 0.022	0.611 ± 0.017	0.743 ± 0.032	0.386 ± 0.014	-0.043 ± 0.054
	NN	0.672 ± 0.027	0.626 ± 0.022	0.753 ± 0.032	0.380 ± 0.018	0.029 ± 0.057
Monomer	RF	0.707 ± 0.019	0.702 ± 0.013	0.810 ± 0.023	0.471 ± 0.008	0.222 ± 0.016
	xGB	0.700 ± 0.014	0.681 ± 0.009	0.800 ± 0.020	0.465 ± 0.011	0.187 ± 0.018
	NN	0.698 ± 0.019	0.672 ± 0.011	0.792 ± 0.022	0.452 ± 0.009	0.208 ± 0.022

Table S7. Performance across 15 dataset splits using embeddings from the encoder of an in-house language model used for reaction prediction.

This performance using these features was much lower than that of all our base models (RF, xGB, and NN) across all chemical featurization types.

f. Binary classification confusion matrices

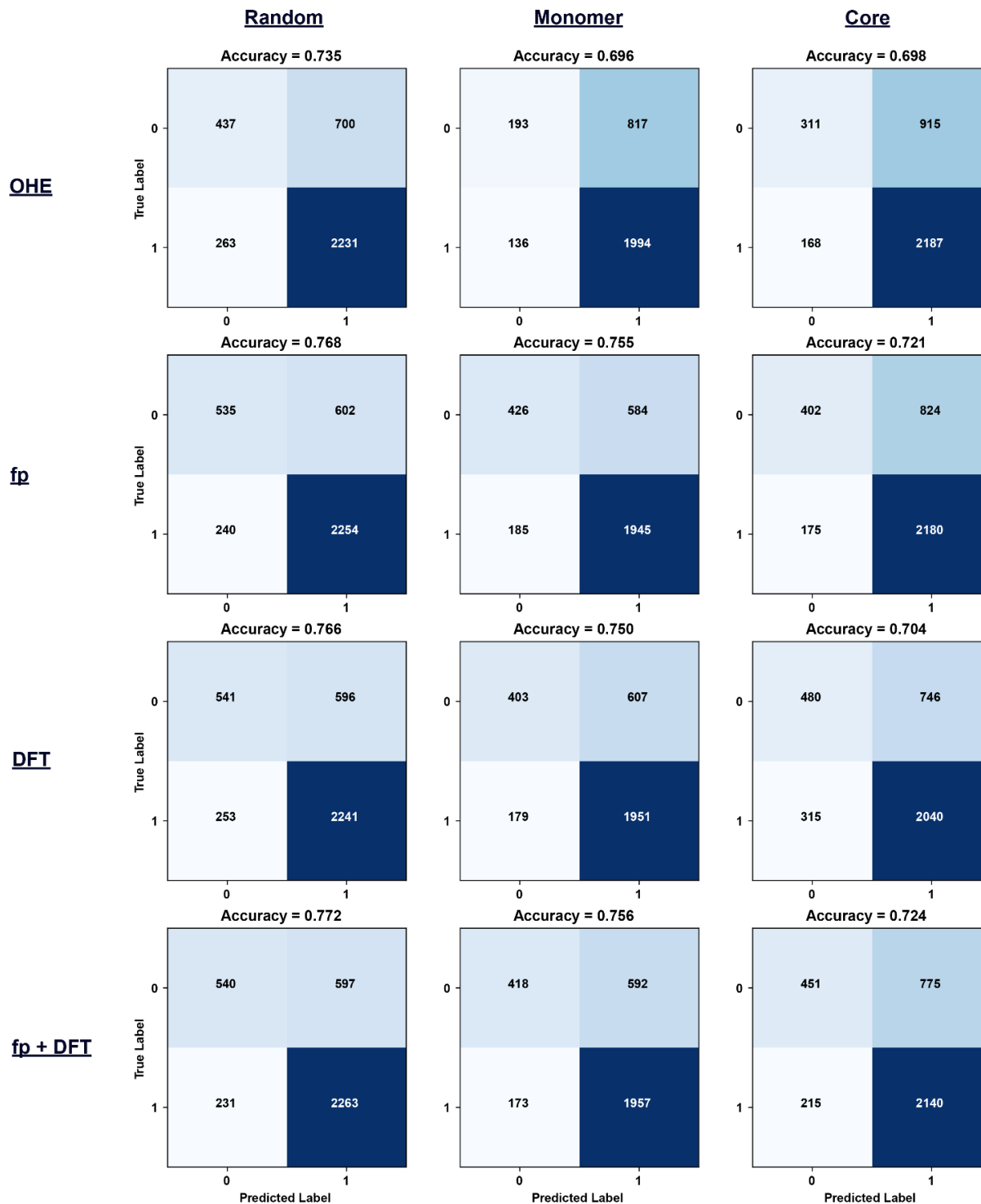


Figure S4. Confusion matrices for the Random Forest Model, evaluated on one of the 15 dataset splits of each type.

g. Multi-class classification confusion matrices

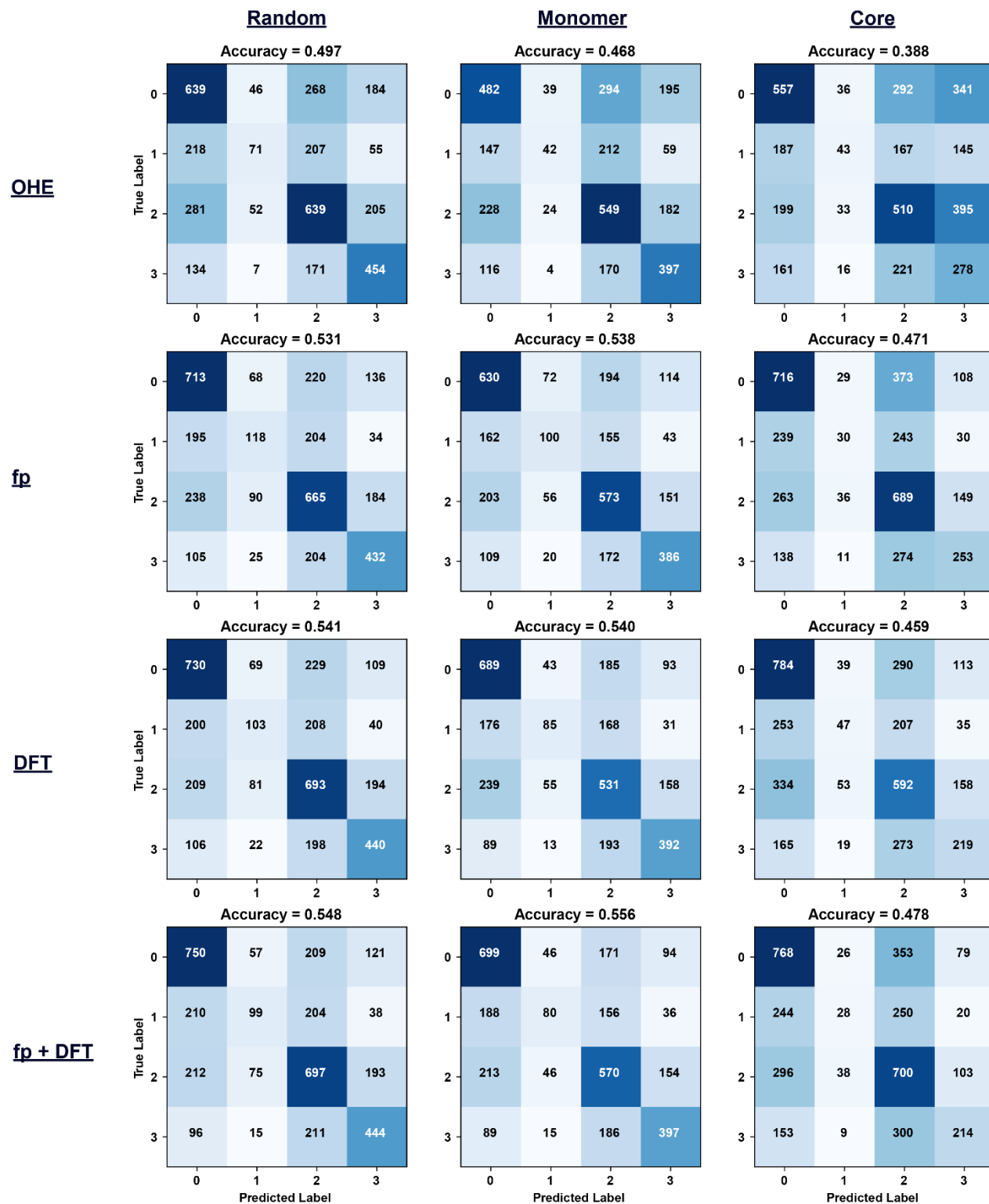


Figure S5. Confusion matrices for the Random Forest Model, evaluated on one of the 15 dataset splits of each type.

h. Regression parity plots

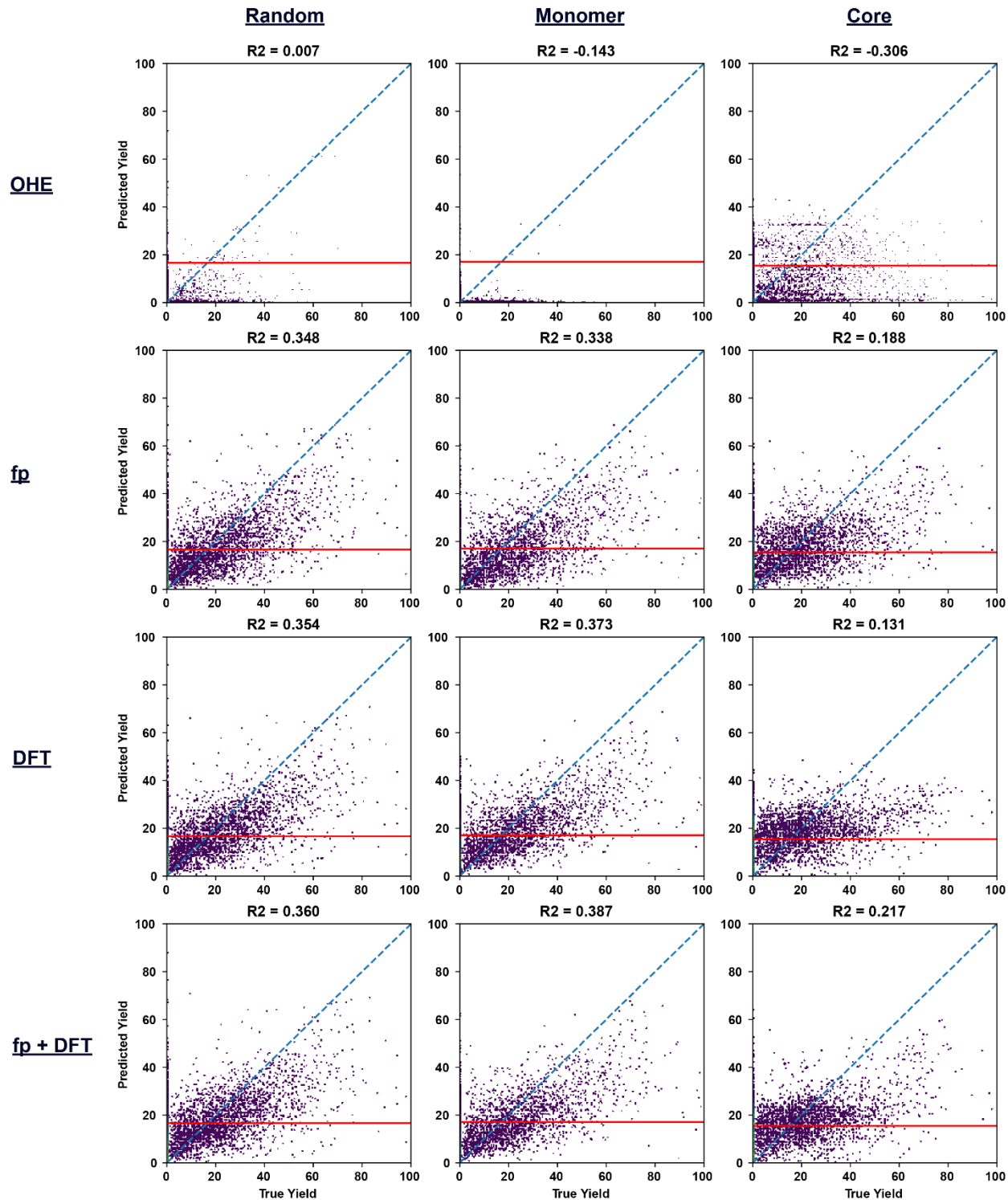


Figure S6. Parity plots for the Random Forest model, evaluated on one test set of the 15 dataset splits of each type. The blue dashed line is the parity line, and the horizontal red line represents the mean percent yield of the test set for each case.

i. Feature importance study

We sought to analyze which features the models were finding as important. We chose to use permutation-based feature importance for this study (via sklearn's `permutation_importance` function), where each feature in the training data is permuted one-by-one, the model trained on the altered data, and the resulting change in performance measured as compared to the originally-trained model with non-permuted features. Specifically, we permuted each feature 5 times to get an average performance drop for that feature, and repeated this across each feature. Theoretically, features that are deemed more important by the model should result in a steeper average performance drop. This procedure is summarized below in Figure S7:

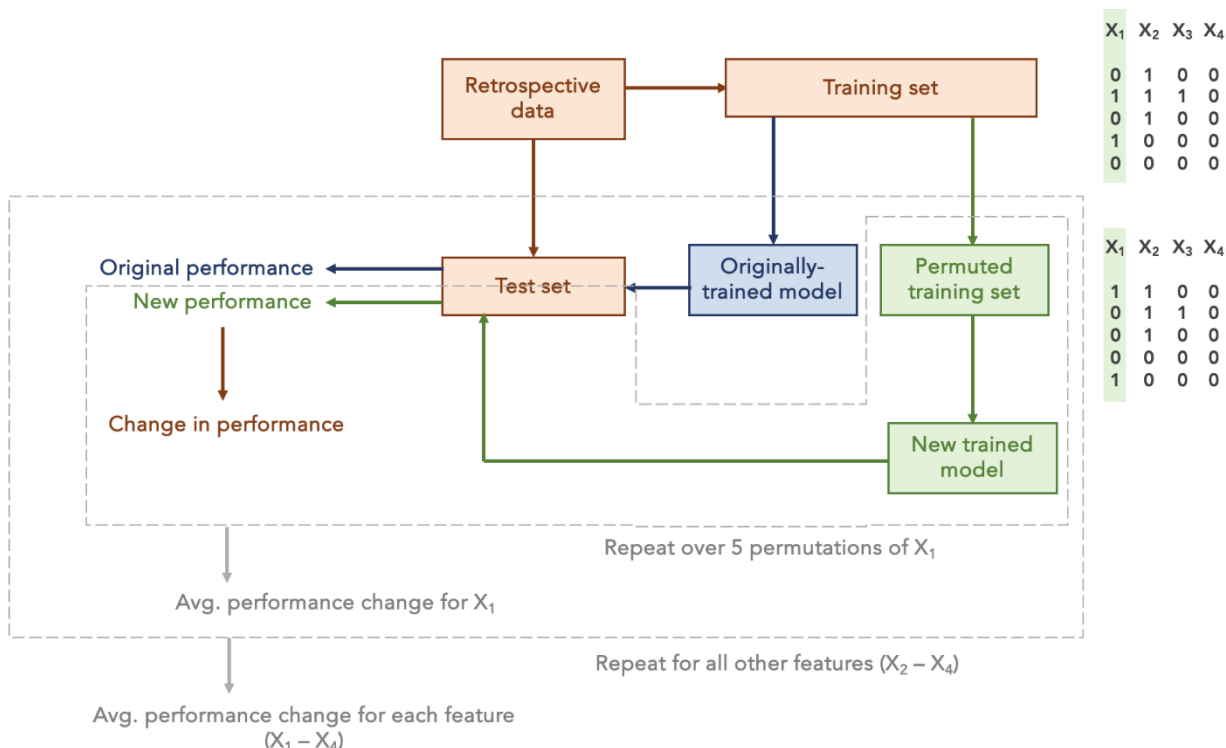


Figure S7. Schematic showing the permutation importance workflow for a single dataset split. The retrospective dataset is split into a training and test set based on the chosen split type and task type (orange), and a model is trained and tested to obtain a performance metric (blue). A feature vector in the training data is then permuted, a new model trained and evaluated, and the resultant performance metric is obtained (green). This is repeated over 5 permutations of the feature vector to obtain an average change in performance for that feature. This procedure is then repeated for all features. In our study, this aggregate process was then repeated for each of the first 5 dataset splits of the chosen split type and task type.

Note the large number of models trained in the overall retrospective study (3 model types x 4 featurization types x 3 task types x 3 split types x 15 splits each = 1620 models total). To re-train each of these models across all these settings, 5 times for each feature in a descriptor set, would have been highly computationally expensive. Therefore, to analyze feature importance we chose to only look at the RF model with fp + DFT features across the first 5 splits of the data for each split type (random/monomer/core), and for each task (binary/multi-class/regression). For each combination of split type and task type (i.e. random split + binary classification), the average composition of the top 50 features across the 5 dataset splits was analyzed. The results for all split and task types are shown below:

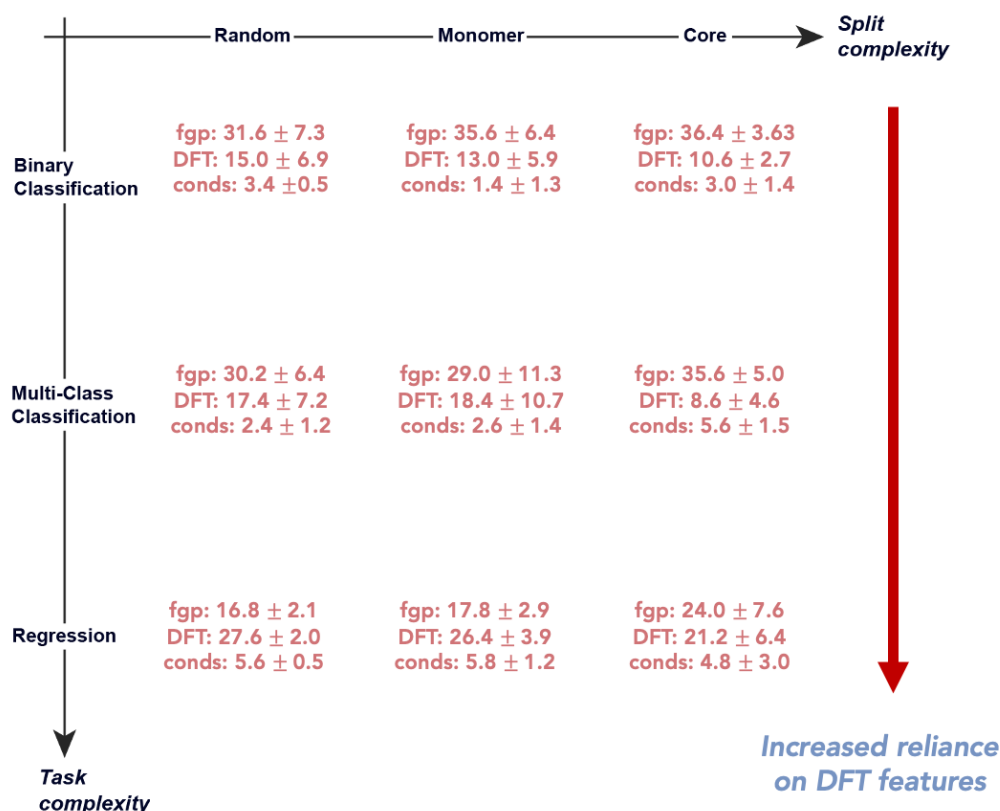


Figure S8. Number of features of each type (fp, DFT, and conditions), amongst the top 50 features for the Random Forest model with fp + DFT featurization, for each combination of split type and task type. Averages and standard deviations are calculated across the top 50 features for the first 5 dataset splits of each type.

These results show a generally increased reliance on DFT features, and decreased reliance on fingerprint features, as the task gets more complex (from binary classification to multi-class classification to regression). We note that the exact features the model deems most important are quite variable, depending on the exact split of the data, but there were a few features that appeared multiple times in the top 50 across different splits and tasks:

- Molecule-level DFT features: boronic acid HOMO energy, aryl halide dipole moment and LUMO energy
- Atom-level DFT features: properties of the boronic acid carbon and boron atoms were considered important, particularly their partial charge and NMR anisotropy
- Structural (fp) (Fig. S9): boronic acid, thiophene, and sulfonamide functional groups; N-containing aromatic heterocycles

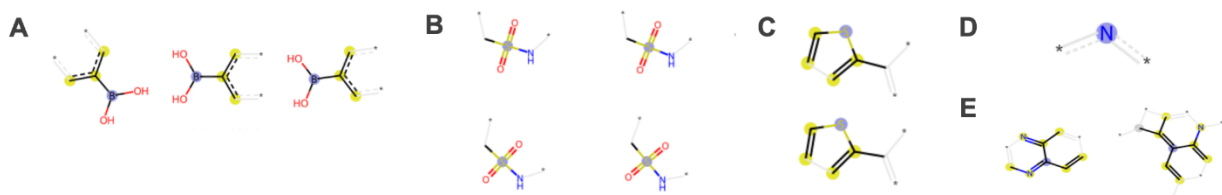


Figure S9. Morgan fingerprint-based substructures, including (A) boronic acid, (B) sulfonamide, and (C) thiophene functional groups; (D) N-containing aromatic heterocycle scaffold, with examples shown in (E). Within each grouping (A/B/C/D/E), each visualization represents the associated substructure in a different molecule from the dataset.

Many of these noted features line up with intuitive patterns that we would hope the model picks up on; for example, with respect to structural features, we observed many instances where a boronic acid monomer furnished a lower average yield than its corresponding boronic ester, which is in alignment with the model generally predicting the former to have lower yields than the latter for the same monomer R-group. We would also expect the model to predict lower yields for substrates with thiophene groups. For the DFT features, partial charges and NMR anisotropy of the reactive-site organoborane atoms would be expected to be important, as these features describe the electronic properties of the atoms that are directly involved in the transmetalation and reductive elimination steps of the reaction. For the molecule-level features, the aryl halide LUMO has been found to be important for interacting with the palladium catalyst, meanwhile the HOMO of the organoborane species has been identified as being important for interacting with the LUMO of the aryl-palladium intermediate prior to transmetalation.⁸

A final note is that due to the number of total features (4096 fp features + 48 DFT features + over 40 conditions features), it is likely that features are highly correlated (particularly for fp features). In other words, the permutation of one feature, even if relatively “important”, is unlikely to cause a noticeable drop in performance, as the model will probably be able to cope by using the other features. Indeed, we found that for each split of the data, the highest performance drop is generally quite small; an example histogram of performance changes for all feature permutations for a core split + binary classification task is shown in Fig. S10 – it is evident that even the highest performance drop is <0.3% in accuracy.

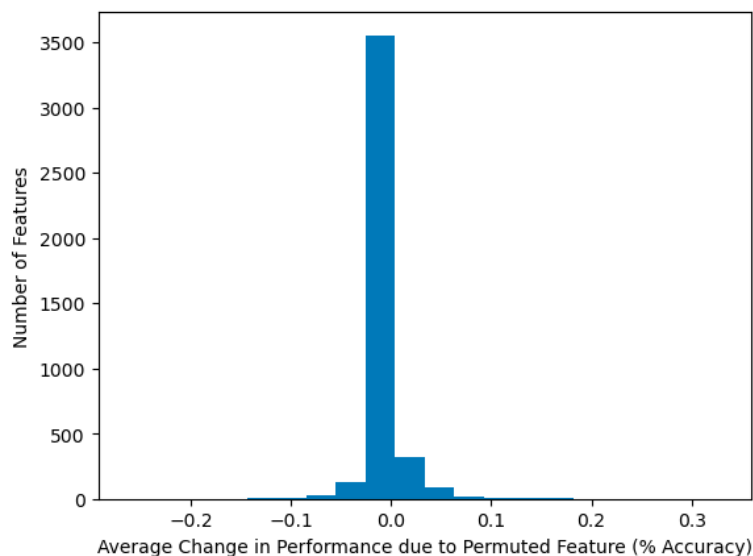


Figure S10. Histogram of average change in performance (as compared to the originally-trained model) when permuting individual features. This plot is for the first core split, binary classification task using the RF model with fp + DFT features.

Because of this, rather than focusing on the precise ordering of importance of the individual features, we have decided to focus on the composition of the top features, as in Fig S8.

j. Post-mid-2021 data: Full performance metrics

	Binary Classification					Multi-Class	Regression
Model	Accuracy	Precision	Recall	AUROC	AUPRC	Accuracy	R ²
RF	0.751	0.797	0.904	0.648	0.839	0.300	-0.003
xGB	0.685	0.794	0.792	0.634	0.840	0.373	-0.045
NN	0.674	0.802	0.760	0.626	0.837	0.345	0.080

Table S8. Performance metrics when testing on the full post-mid-2021 dataset. All models shown used concatenated FP + DFT features.

			Binary Classification			Multi-Class	Regression		
Core	Monomer	Model	Accuracy	AUROC	AUPRC	Accuracy	R ²	MAE	RMSE
Seen	Seen	RF	0.800	0.684	0.869	0.309	-0.048	0.169	0.212
		xGB	0.782	0.665	0.882	0.446	0.027	0.161	0.204
		NN	0.700	0.644	0.887	0.346	0.011	0.158	0.206
Seen	Unseen	RF	0.671	0.592	0.748	0.329	-0.330	0.149	0.172
		xGB	0.536	0.509	0.728	0.336	-0.299	0.140	0.170
		NN	0.471	0.519	0.705	0.336	-0.039	0.118	0.152
Unseen	Seen	RF	0.772	0.644	0.856	0.317	-0.057	0.211	0.261
		xGB	0.738	0.648	0.853	0.359	-0.138	0.220	0.270
		NN	0.768	0.644	0.845	0.342	0.041	0.197	0.248
Unseen	Unseen	RF	0.759	0.646	0.872	0.195	-0.104	0.198	0.248
		xGB	0.655	0.615	0.857	0.379	-0.199	0.207	0.259
		NN	0.713	0.630	0.880	0.368	-0.130	0.199	0.251

Table S9. Performance metrics for the 4 data subsets of the post-mid-2021 dataset. All models shown used concatenated FP + DFT features.

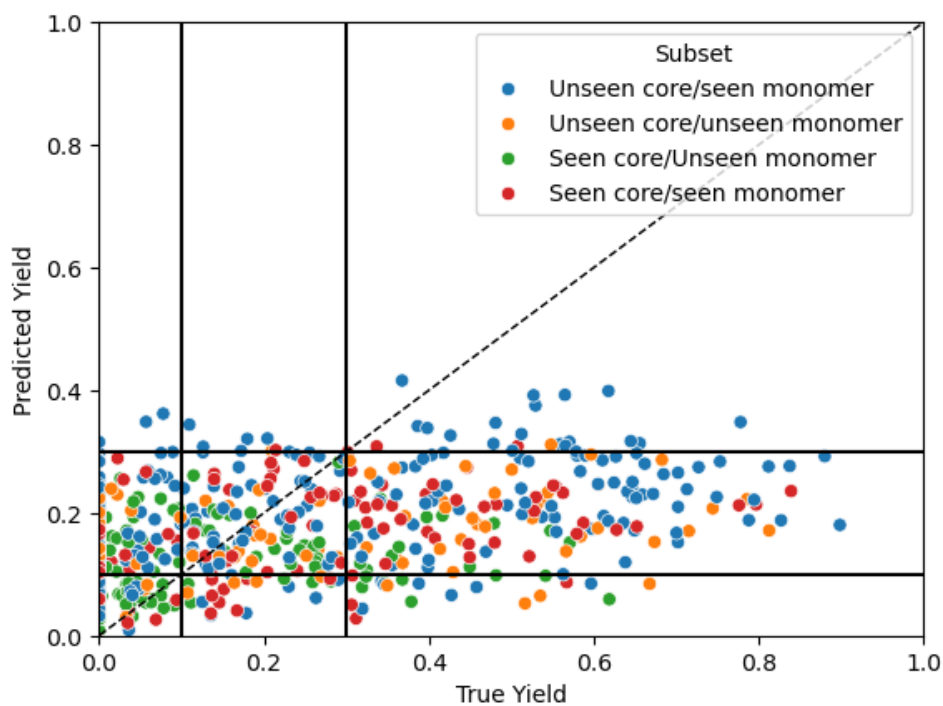


Figure S11. Parity plot showing the neural network’s predictions on the post-2021 data as compared to the actual yields and colored by the 4 subsets mentioned in the main paper. Cutoff lines have been placed on the figure at 10% and 30% true and predicted yield.

3. Supplementary Figures

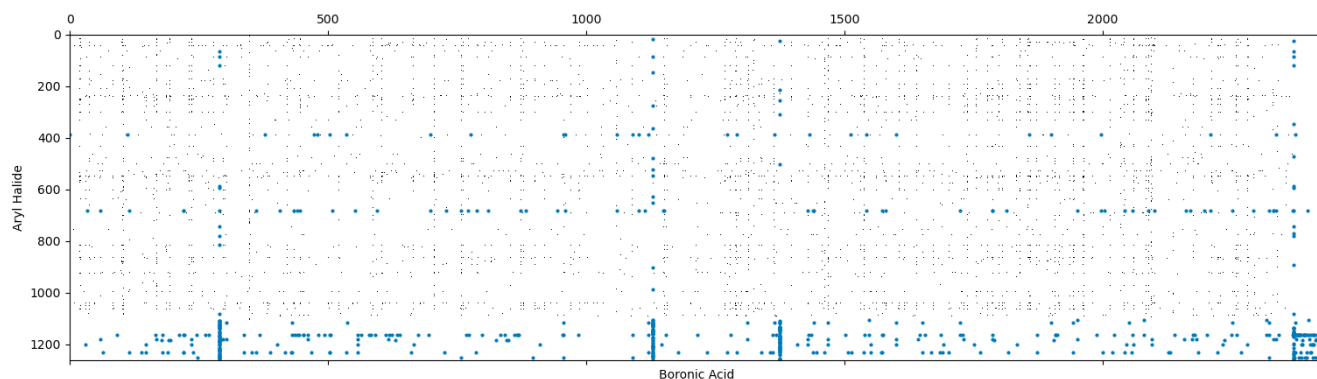


Figure S12. Sparsity plot showing the combinations of aryl halide and boronic acid present in the datasets vs. all possible combinations of unique aryl halides and boronic acids from the data. Conditions not included. The black points represent a measured combination from the retrospective dataset, while the larger blue points represented a measured combination from the post-2021 dataset.

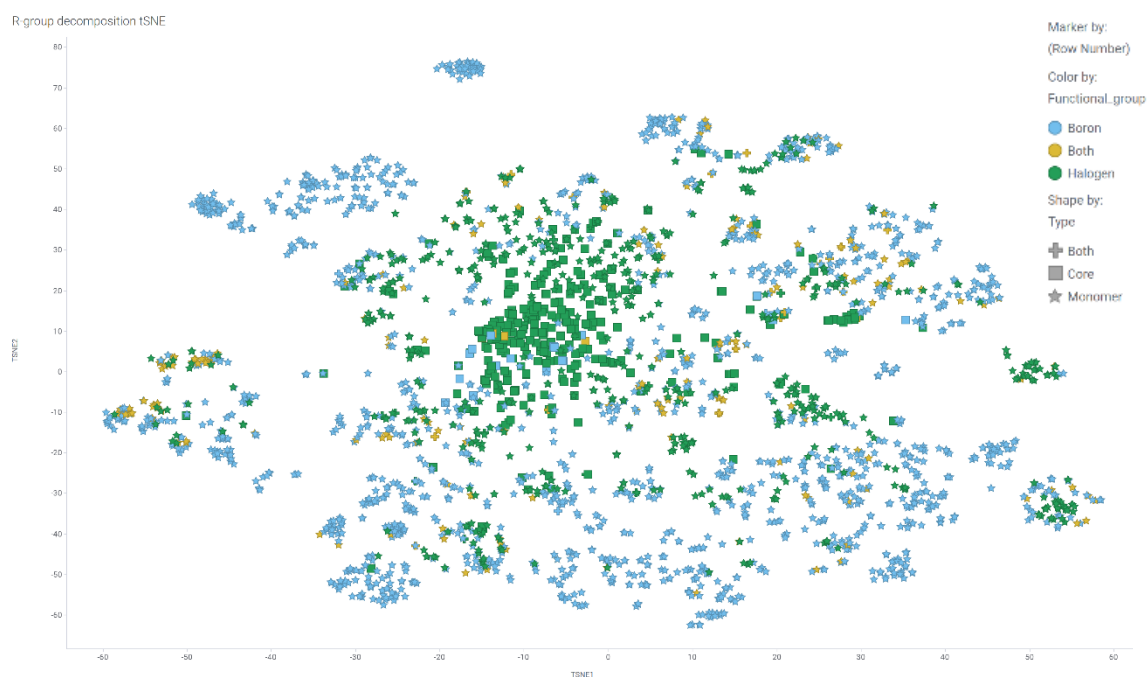


Figure S13. tSNE plot of the decomposed core and monomer R-groups found in the retrospective dataset. The plot was generated via tSNE visualization of the PCA decomposition of the R-groups' MinHash fingerprints.⁷

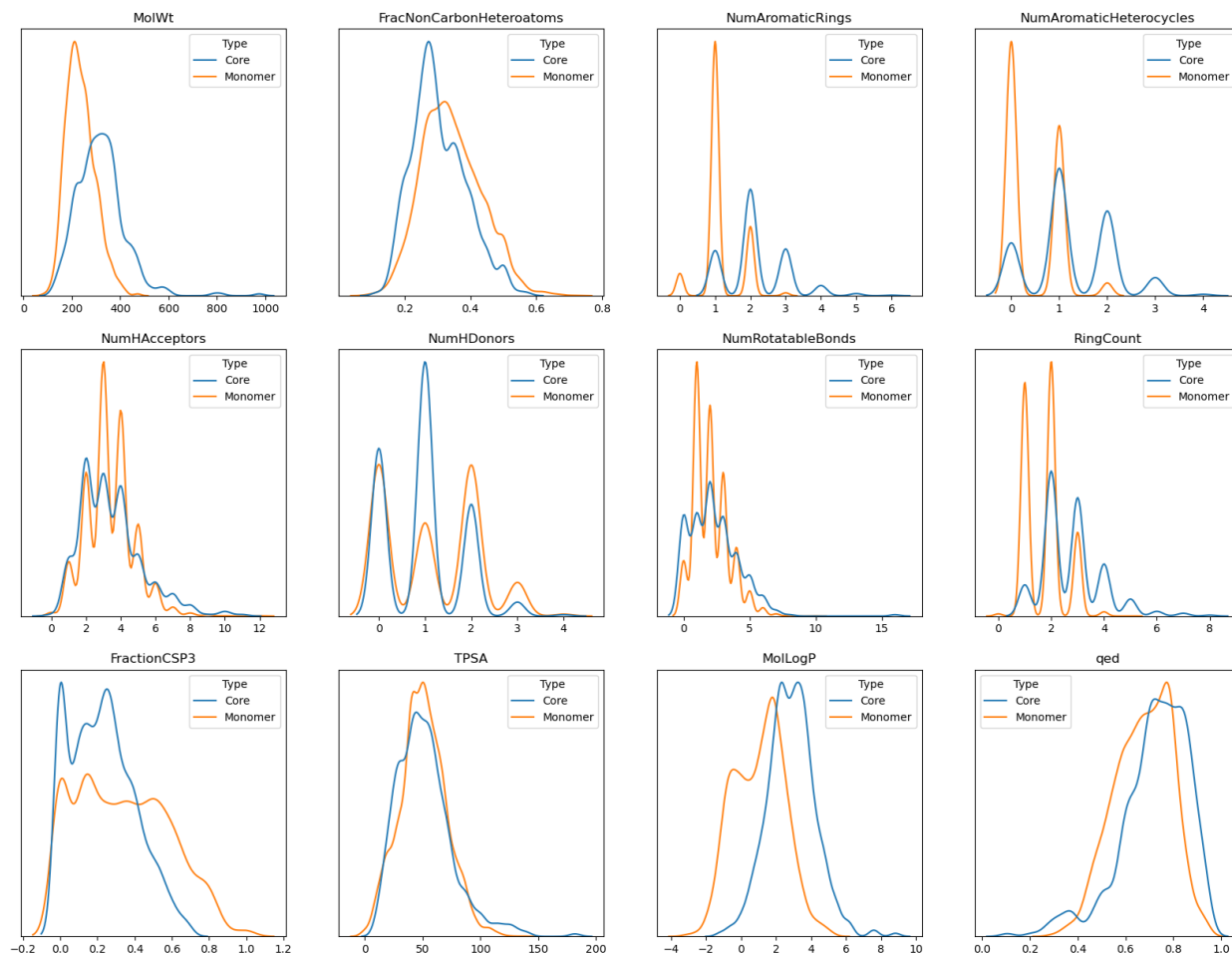


Figure S14. KDE (Kernel Density Estimation) plots of various physiochemical properties across the structures in the retrospective dataset. Note that the core and monomer distributions have been normalized separately for each property.

Properties shown include MolWt (molecular weight), FracNonCarbonHeteroatoms (# of non-C heteroatoms/# of heavy atoms), NumAromaticRings (# of aromatic rings), NumAromaticHeterocycles (# of aromatic heterocycles), NumHAcceptors (# of H-bond acceptors), NumHDonors (# of H-bond donors), NumRotatableBonds (# of rotatable bonds), RingCount (ring count), FractionCSP3 (fraction of sp³ C atoms), TPSA (topological polar surface area), MolLogP (Crippen's computational LogP estimation),⁹ and qed (quantitative estimate of druglikeness). All properties were computed via RDKit's Descriptors module.

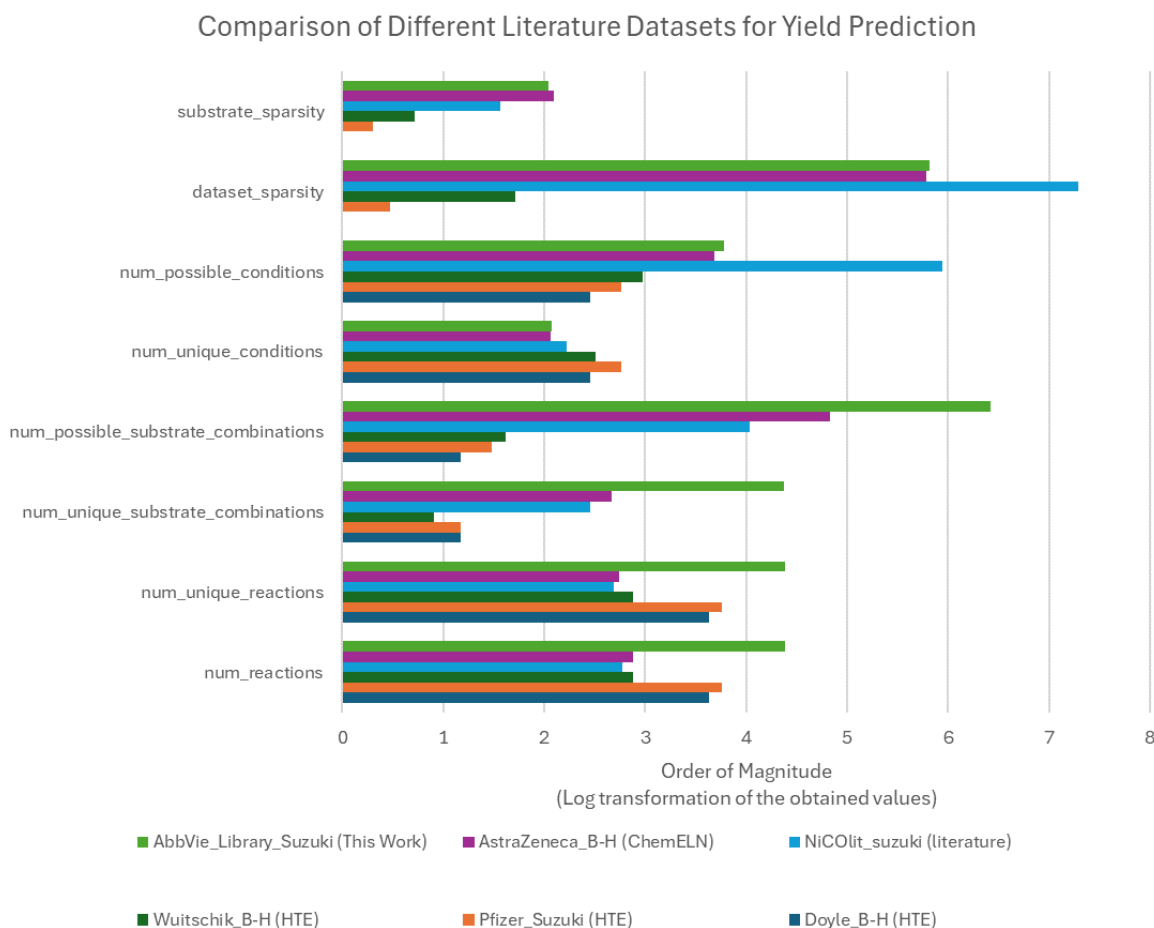


Figure S15. Comparison of AbbVie’s Suzuki library dataset (top bar in all cases) to other datasets used for ML yield prediction. The datasets were compared in terms of the number of unique conditions and substrate pairs seen within the data, as well as the number of total possible combinations of conditions and substrates. Dataset sparsity is defined as the total number of measured reactions divided by the product of the total possible number of reactant pairs and conditions, while substrate sparsity is defined as the number of unique reactant pairs divided by the total possible number of reactant pairs (note: these values were negative-log transformed and therefore larger bars indicate more sparsity). Note: B-H stands for the Buchwald-Hartwig reaction.

Other datasets used (in descending order): AstraZeneca Buchwald-Hartwig,¹⁰ NiCOLit (Suzuki reactions),¹¹ Wuitschik Buchwald-Hartwig,¹² Pfizer Suzuki,¹³ Doyle Buchwald-Hartwig.¹⁴

AbbVie’s Suzuki library dataset has about an order of magnitude larger number of reactions than the other datasets, and exhibits more unique substrate pairs, while maintaining a similar number of unique conditions as the smaller datasets. Therefore, we hypothesized that our parallel library dataset would have a better chance of extrapolating to unseen substrate pairs.

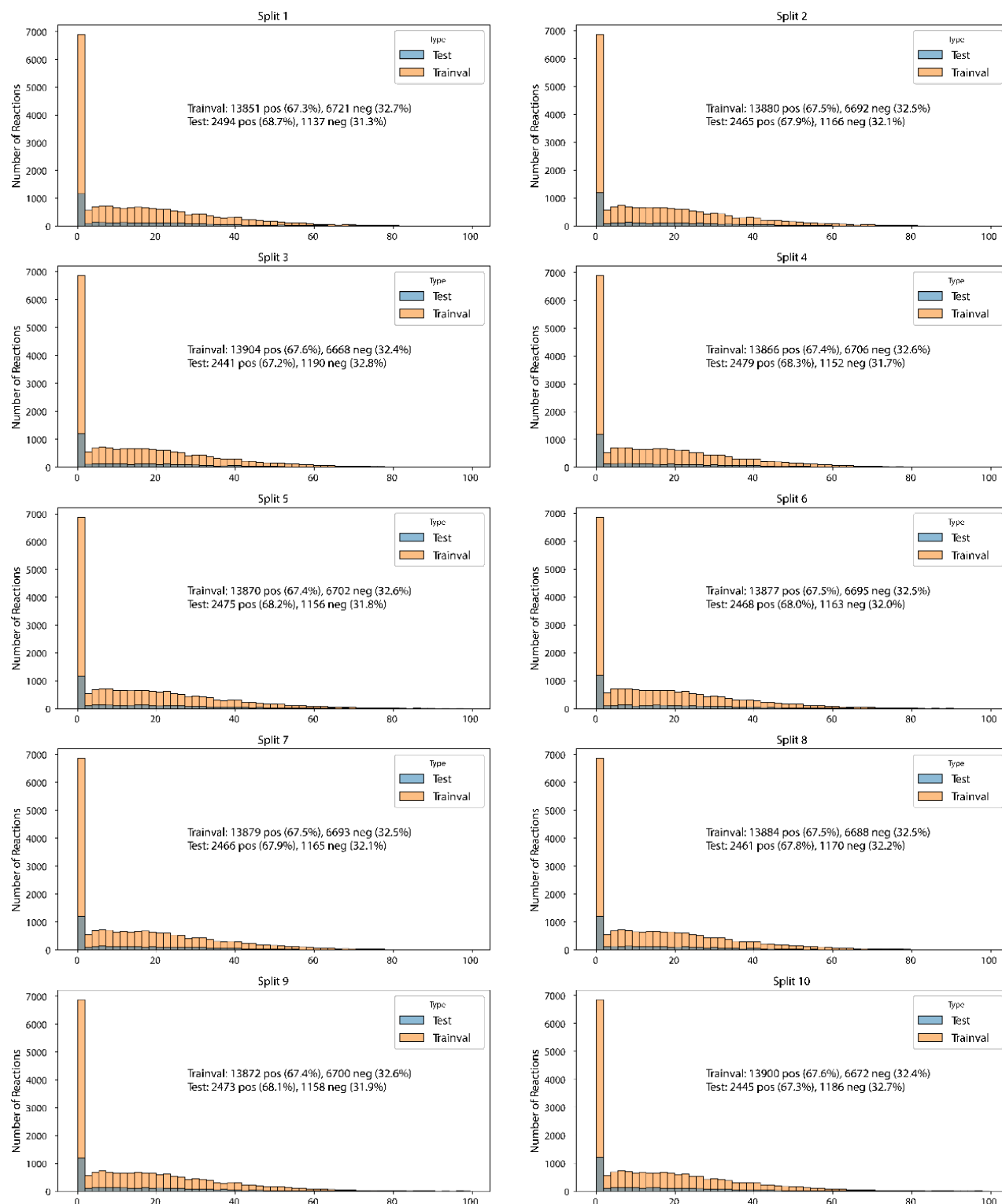


Figure S16a. Yield histograms of the combined train and validation (“trainval”) and test sets across random splits 1-10 of the retrosynthetic dataset. The counts and percentage of positive (>0 yield) and negative (0-yield) samples within each trainval and test set are annotated on each plot.

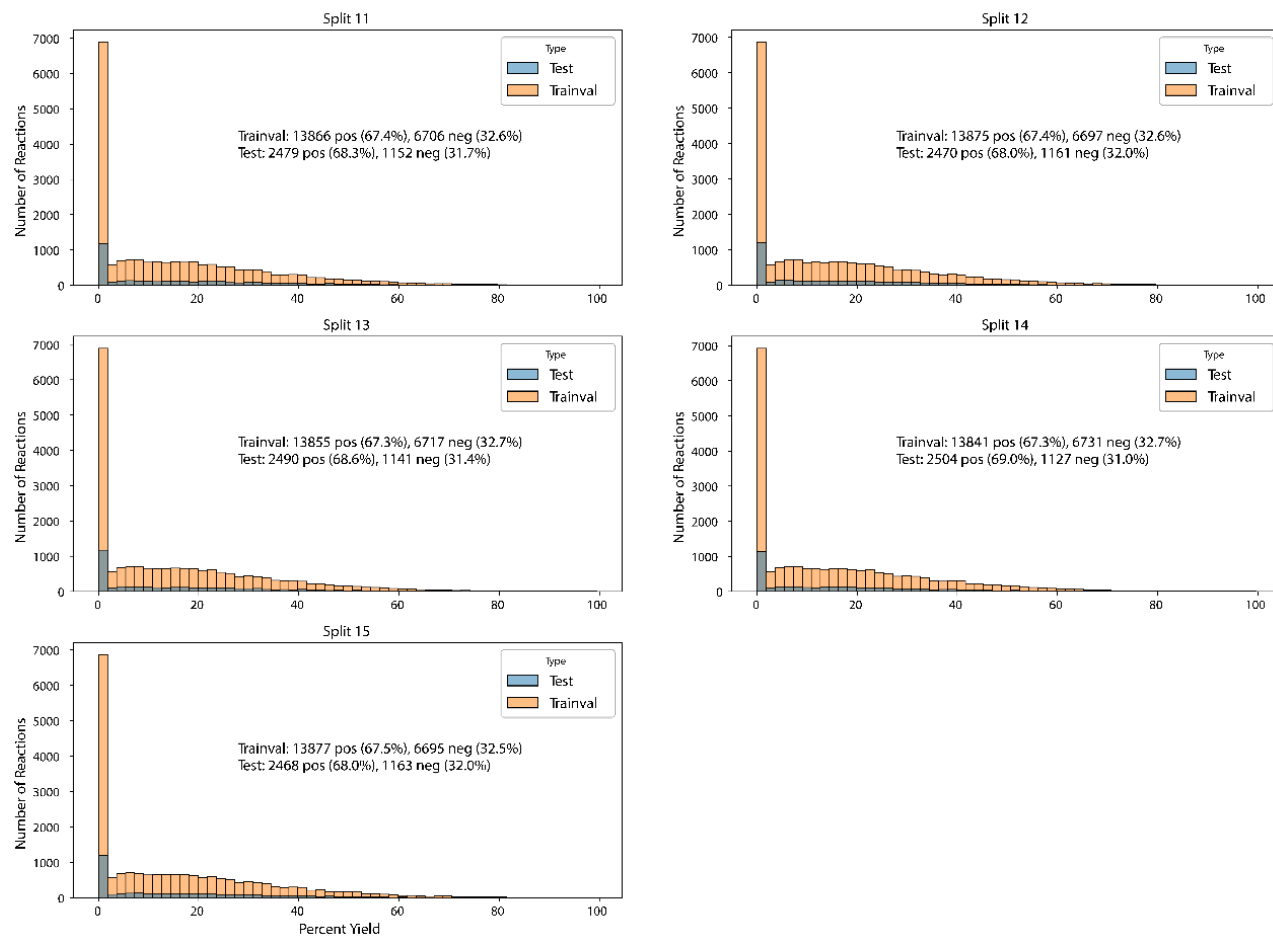


Figure S16b. Yield histograms of the combined train and validation (“trainval”) and test sets across random splits 11-15 of the retrosynthetic dataset. The counts and percentage of positive (>0 yield) and negative (0-yield) samples within each trainval and test set are annotated on each plot.

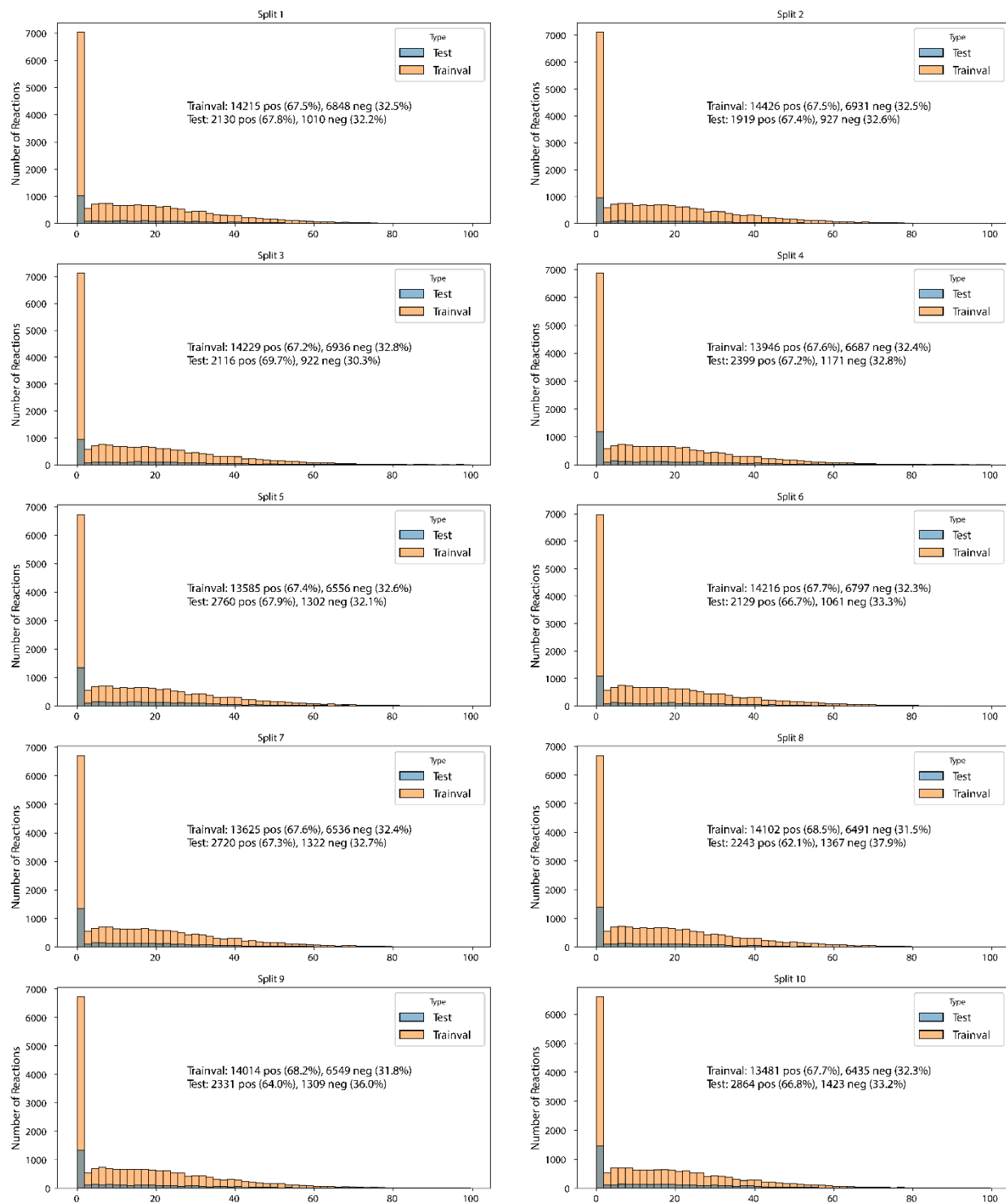


Figure S17a. Yield histograms of the combined train and validation (“trainval”) and test sets across monomer splits 1-10 of the retrospective dataset. The counts and percentage of positive (>0 yield) and negative (0-yield) samples within each trainval and test set are annotated on each plot.

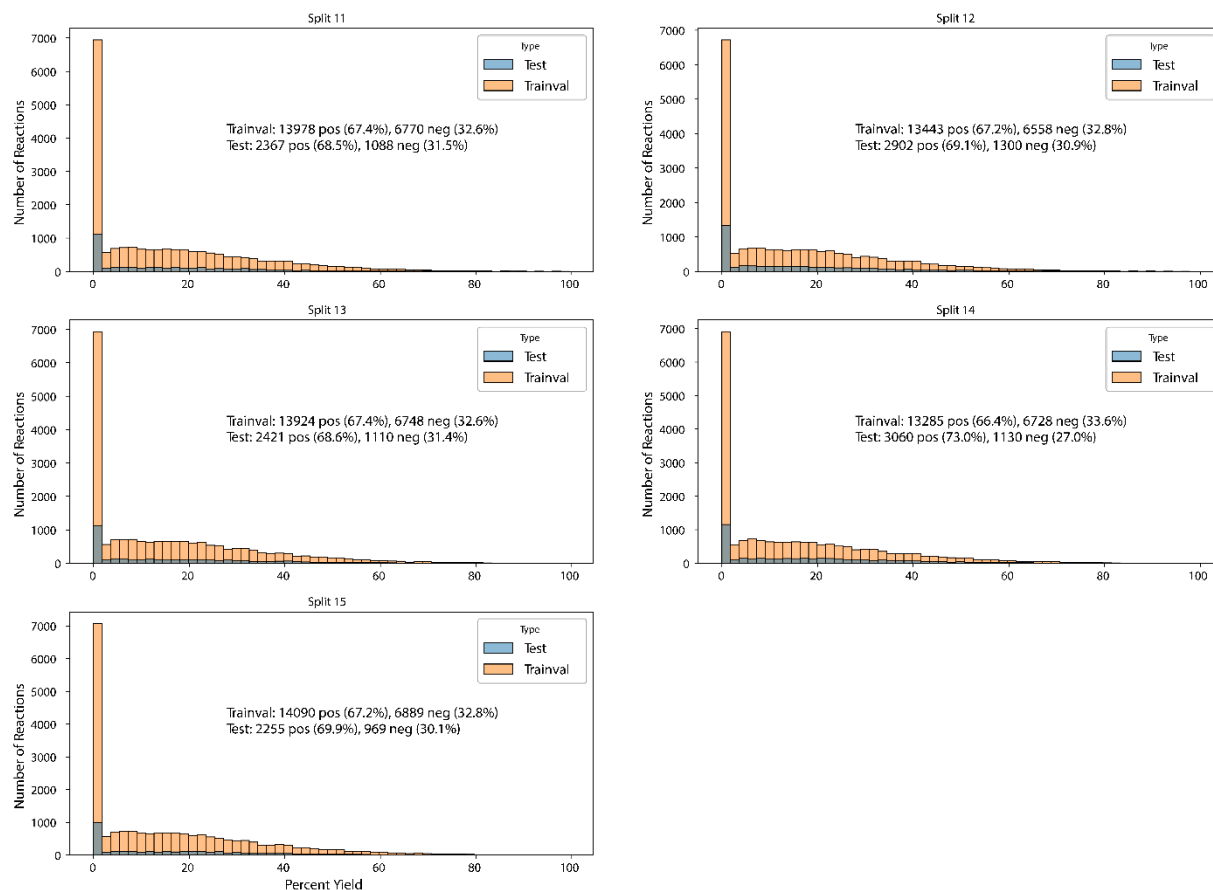


Figure S17b. Yield histograms of the combined train and validation ("trainval") and test sets across monomer splits 11-15 of the retrospective dataset. The counts and percentage of positive (>0 yield) and negative (0-yield) samples within each trainval and test set are annotated on each plot.

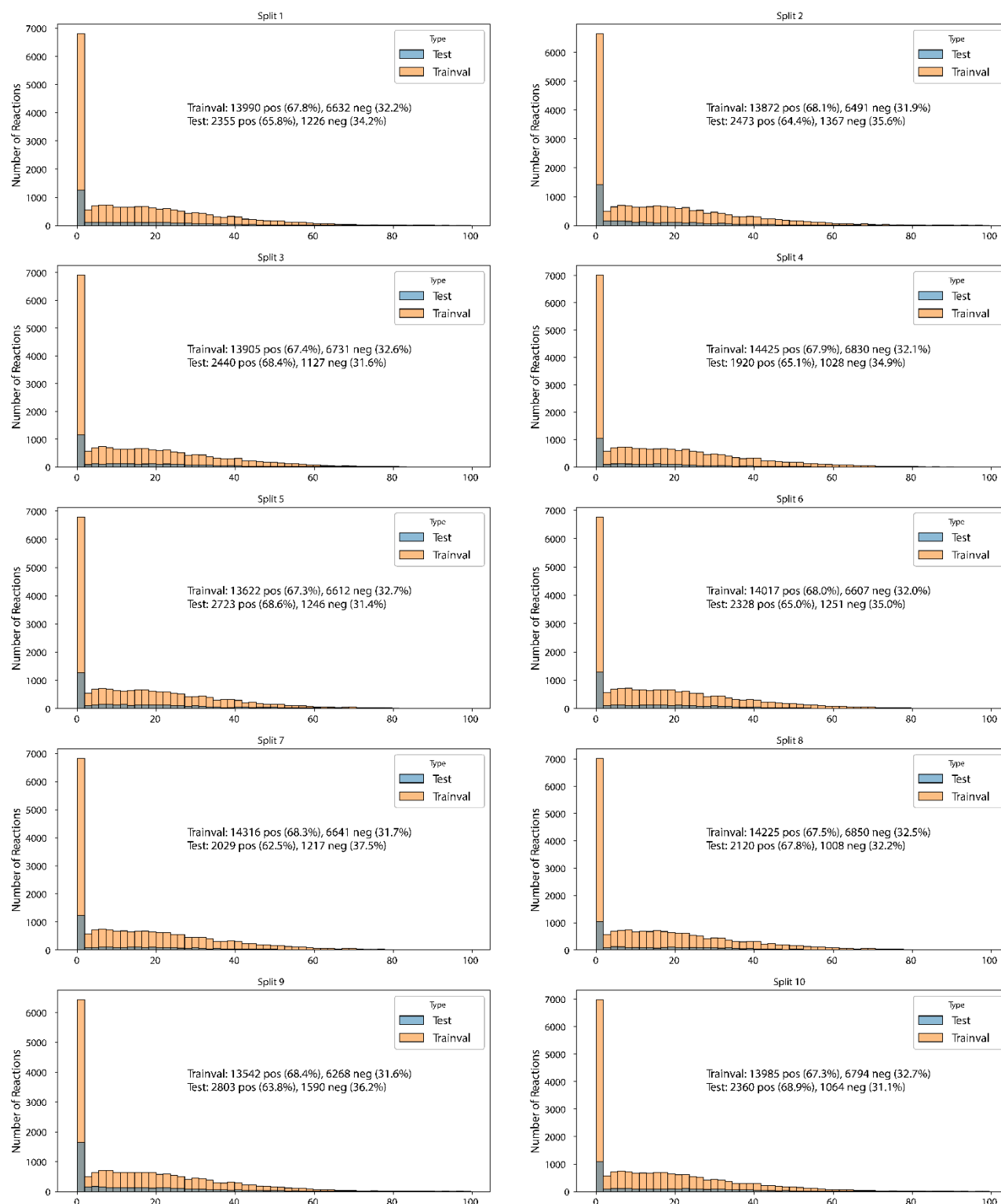


Figure S18a. Yield histograms of the combined train and validation (“trainval”) and test sets across core splits 1-10 of the retrosynthetic dataset. The counts and percentage of positive (>0 yield) and negative (0-yield) samples within each trainval and test set are annotated on each plot.

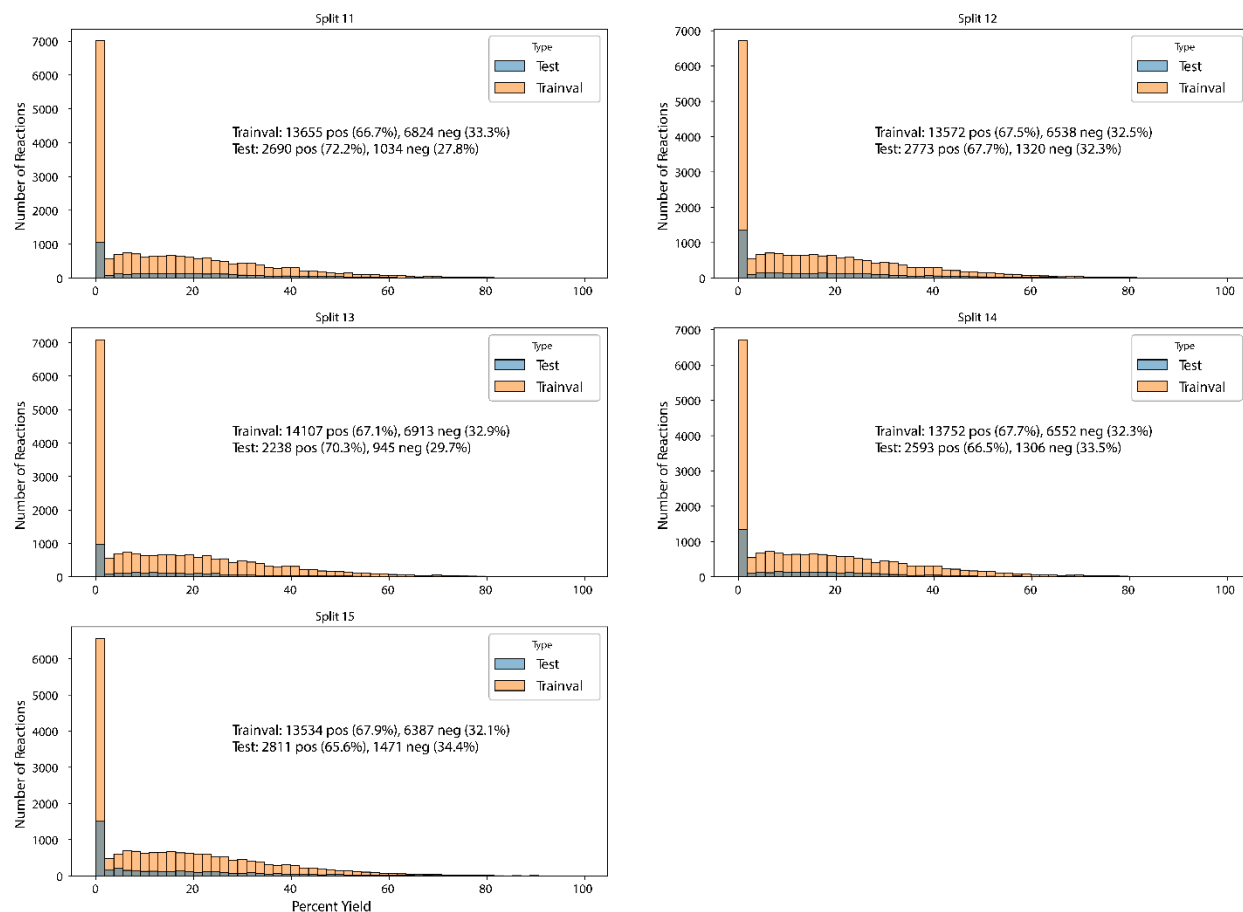


Figure S18b. Yield histograms of the combined train and validation (“trainval”) and test sets across core splits 11-15 of the retrospective dataset. The counts and percentage of positive (>0 yield) and negative (0-yield) samples within each trainval and test set are annotated on each plot.

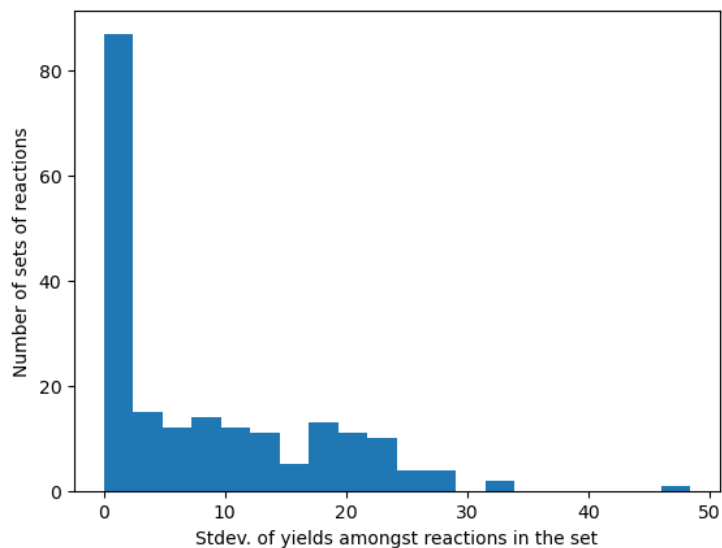


Figure S19. Histogram showing the standard deviation of yields amongst sets of “repeated” reactions (same aryl halide, boronic acid, catalyst, ligand, base, and solvent) in the retrospective dataset. In total, there were 201 sets of repeated reactions, 200 of which contained 2 repeated reactions, and 1 of which contained 3, for a total of 403 repeated reactions. Over half of the sets have a very low standard deviation of yields (less than 5%). This analysis was done in an attempt to quantify dataset noise, although it must be noted that these repeated reactions could still have differences not captured by our data, for example other reaction/analysis conditions such as temperature, time, or purification gradient.

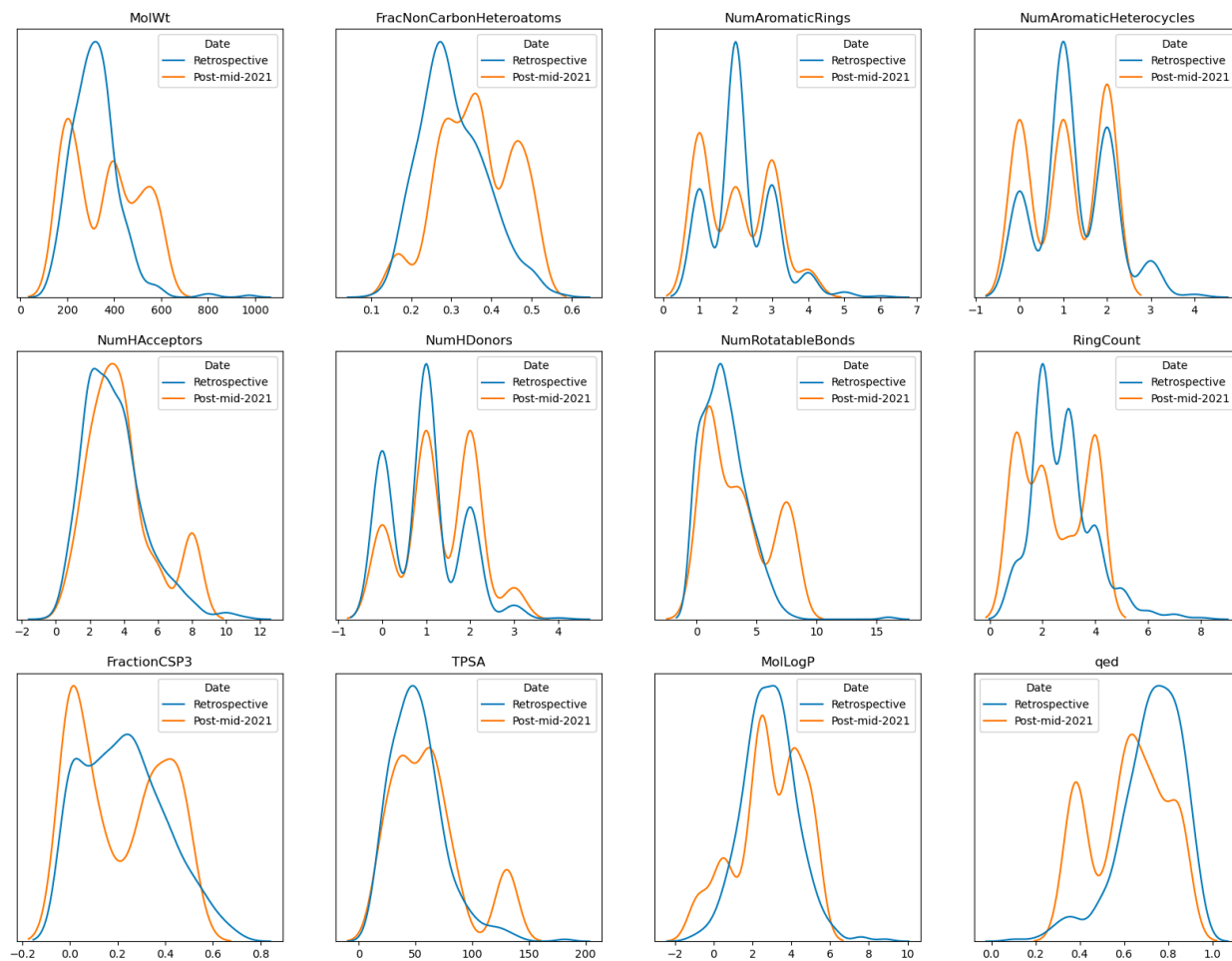


Figure S20. KDE (Kernel density estimation) plots showing distributions of the core structures in the retrospective and post-mid-2021 datasets across various physiochemical properties important for compound design. Note that the retrospective and post-mid-2021 distributions have been normalized separately for each property. See the discussion under Figure S14 for explanations of each property.

The post-mid-2021 core structures exhibit, on average, higher molecular weights, more non-C heteroatoms, more rotatable bonds, and lower qed. This suggests the increased complexity of the post-mid-2021 cores as compared to the retrospective cores.

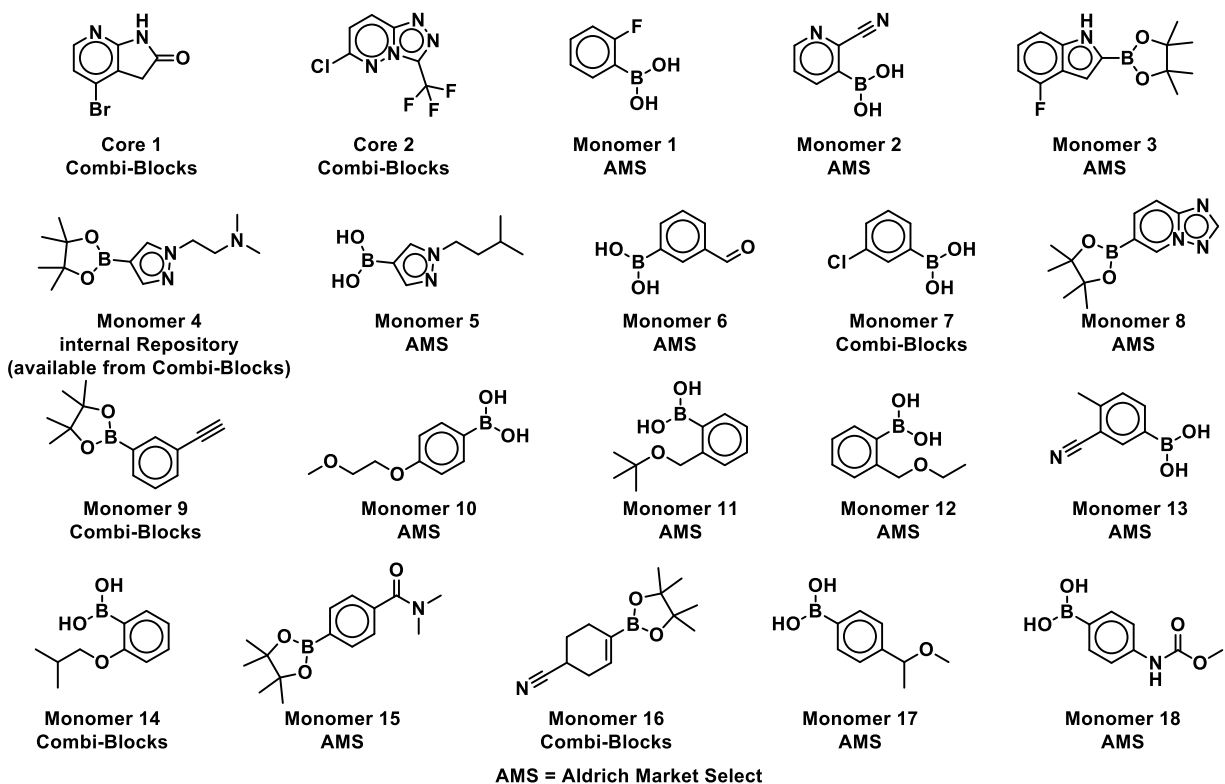


Figure S21. Structures of Cores and Monomers Used in the Monomer Replacement Experiments

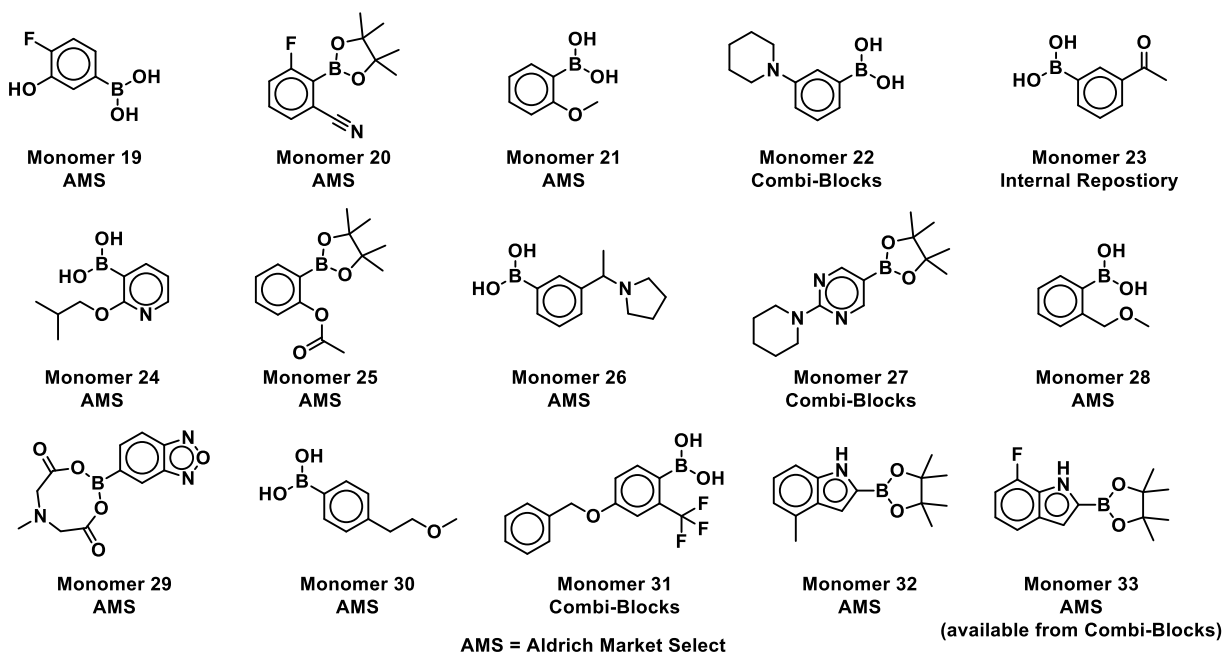
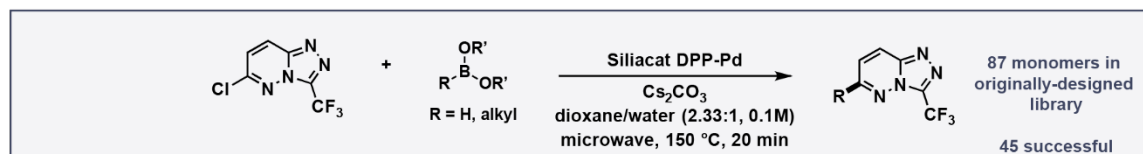


Figure S22. Structures of Monomers Used in the Monomer Replacement Experiments (cont'd)



monomer rescue: using the ML model after conducting the library

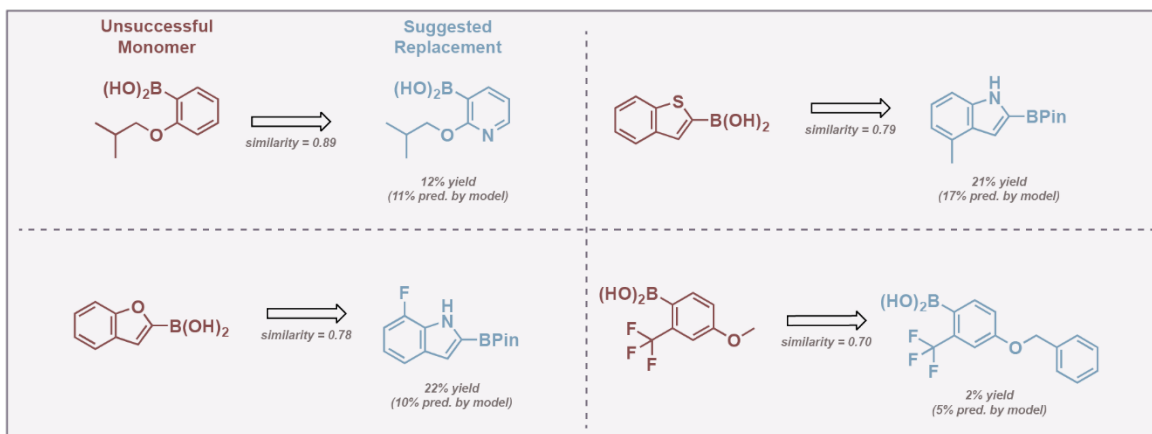


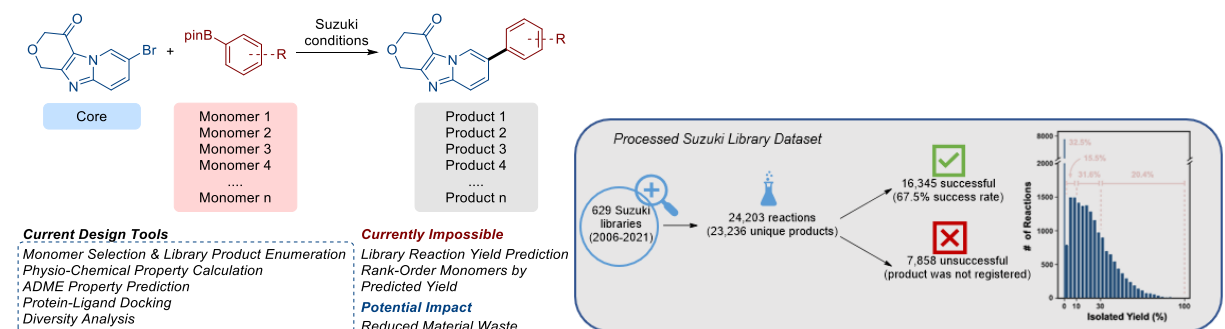
Figure S23. Further Demonstrating Monomer Rescue with a Different Core

4. Chemist Comparison Study Details

Predictions made by the models were compared to those made by 11 expert medicinal chemists with varying backgrounds and years of experience. Six libraries were selected for evaluation for this study, 3 of which were whole-library predictions (compared to the core-split model's predictions), and the remaining 3 being partial-library predictions (compared to the monomer-split model's predictions). The medicinal chemists were provided with directions (shown below), along with two Excel files showing the full structures of the cores and monomers present in the evaluated libraries with their reaction conditions. The chemists entered their predictions as multi-class yield bins, and their predictions for reaction success and binned yields were compared to the RF (FP+DFT, default hyperparameters) classification models' predictions (binary and multi-class). One reaction from **Library 5** was excluded from the comparison, as the monomer used in the reaction appeared as a core in the retrospective dataset and therefore, was removed from the comparison due to the model previously encountering the structure during training.

Document provided to the chemists:

Background: The ACT group has conducted >20,000 single-step Suzuki library reactions since 2006. Current library design tools allow for product property calculation/prediction, protein-ligand docking, diversity analysis, etc. to filter a set of monomers. It is not yet possible to use machine learning (ML) to predict the yield of a library reaction, though, and this would significantly aid in the design process by reducing material loss to failed reactions. As such, we sought to evaluate our large Suzuki library dataset to build a yield prediction function into the library design workflow. Overall, this dataset exhibits a success rate of 68%, where “success” is defined as the product being registered in the A-room (success rates for individual libraries may vary, though). Furthermore, we divide the successful reactions into three yield bins: 0-10%, 10-30%, and >30%. Statistics for the entire dataset are shown below:



Purpose: The purpose of this exercise is to compare the predictions made by our ML model to those made by expert medicinal chemists. Two scenarios will be covered: the prediction of entire libraries, and the prediction of select monomers from libraries. In the latter scenario, yields will be provided for the non-testing monomers from the library.

Directions: Open the library files with Excel (please do not delete any columns!). Use your experience as a medicinal chemist to predict whether the reactions in the libraries will be successful. Please refrain from using outside references and use only your experience and chemical intuition. For each reaction, indicate your predicted yield by entering the following numbers (see below for an example):

Catalyst	Monomer	Ligand	Solvent	Synthesis	Monomer_Conc'd	Yield
[PA Pd G3] [C3PO4]	[None]	[Dioxane]		5123457		
[PA Pd G3] [C3PO4]	[None]	[Dioxane]		5123458		
[PA Pd G3] [C3PO4]	[None]	[Dioxane]		5123460		
[PA Pd G3] [C3PO4]	[None]	[Dioxane]		5123461		
[PA Pd G3] [C3PO4]	[None]	[Dioxane]		5123462		

- 0 – fail/product could not be registered
- 1 – product was registered with 0.1-10% yield
- 2 – product was registered with 10.1-30% yield
- 3 – product was registered with >30% yield

	Chemist 1	Chemist 2	Chemist 3	Chemist 4	Chemist 5	Chemist 6	Chemist 7	Chemist 8	Chemist 9	Chemist 10	Chemist 11	Chemist Average	Binary Model	Difference
Whole-Library	0.7	0.68	0.7	0.7	0.73	0.76	0.75	0.72	0.73	0.7	0.7	0.72	0.8	0.08
Partial-Library	0.57	0.59	0.7	0.72	0.67	0.72	0.72	0.63	0.61	0.67	0.65	0.66	0.76	0.10

Table S10. Split-Level Performance of Individual Chemists vs the binary classification models for predicting reaction success

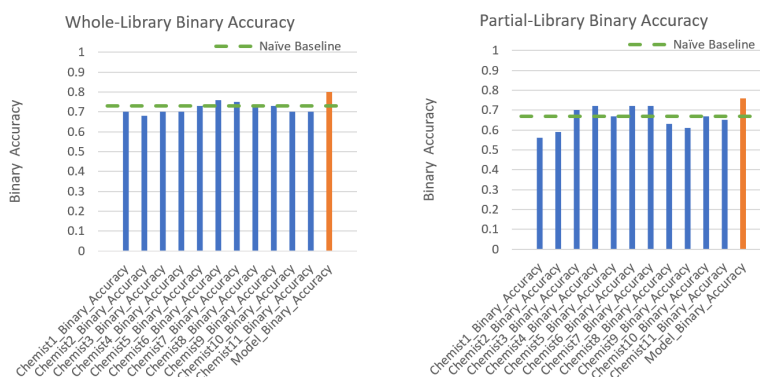


Figure S24. Split-Level Performance of Individual Chemists vs the binary classification models for predicting reaction success. Blue represents the chemists while orange represents the model. The naïve baseline (class = 1) is drawn for each library as a horizontal dashed green line.

	Chemist 1	Chemist 2	Chemist 3	Chemist 4	Chemist 5	Chemist 6	Chemist 7	Chemist 8	Chemist 9	Chemist 10	Chemist 11	Chemist Average	Multi-Class Model	Difference
Whole-Library	0.17	0.54	0.42	0.15	0.25	0.27	0.52	0.37	0.23	0.41	0.48	0.35	0.59	0.24
Partial-Library	0.43	0.46	0.5	0.35	0.52	0.46	0.46	0.46	0.22	0.57	0.39	0.44	0.59	0.15

Table S11. Split-Level Performance of Individual Chemists vs the multi-class classification models for predicting binned reaction yields.

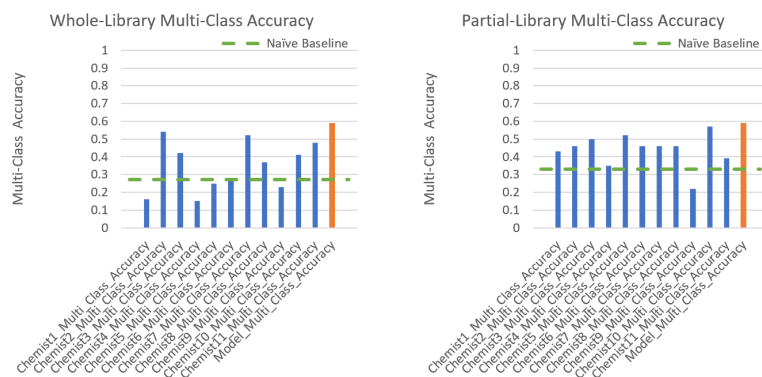


Figure S25. Split-Level Performance of Individual Chemists vs the multi-class classification models for predicting binned reaction yields. Blue represents the chemist while orange represents the model. The naïve baseline (class = 0) is drawn for each library as a horizontal dashed green line.

	Chemist 1	Chemist 2	Chemist 3	Chemist 4	Chemist 5	Chemist 6	Chemist 7	Chemist 8	Chemist 9	Chemist 10	Chemist 11	Chemist Average	Binary Model	Naive Baseline
Library 1	0.76	0.61	0.79	0.73	0.73	0.73	0.76	0.82	0.73	0.73	0.70	0.73	0.88	0.76
Library 2	0.56	0.72	0.50	0.67	0.61	0.78	0.72	0.67	0.67	0.61	0.67	0.65	0.56	0.67
Library 3	0.75	0.75	0.75	0.70	0.85	0.80	0.75	0.60	0.80	0.75	0.75	0.75	0.90	0.75
Library 4	0.48	0.57	0.61	0.65	0.57	0.70	0.61	0.57	0.52	0.65	0.52	0.58	0.74	0.48
Library 5	0.91	0.91	0.91	0.91	1.00	1.00	1.00	0.73	0.91	1.00	0.91	0.93	1.00	1.00
Library 3	0.42	0.33	0.67	0.67	0.58	0.50	0.67	0.67	0.50	0.42	0.67	0.55	0.58	0.75

Table S12. Library-Level Performance of Individual Chemists vs the binary classification models for predicting reaction success.



Figure S26. Library-Level Performance of Individual Chemists vs the binary classification models for predicting reaction success. Blue represents the chemists while orange represents the model. The naïve baseline (class = 1) is drawn for each library as a horizontal dashed green line.

	Chemist 1	Chemist 2	Chemist 3	Chemist 4	Chemist 5	Chemist 6	Chemist 7	Chemist 8	Chemist 9	Chemist 10	Chemist 11	Chemist Average	Multi-Class Model	Naive Baseline
Library 1	0.21	0.52	0.64	0.15	0.18	0.30	0.52	0.48	0.21	0.39	0.48	0.37	0.73	0.24
Library 2	0.17	0.56	0.11	0.28	0.28	0.33	0.44	0.33	0.28	0.44	0.33	0.32	0.33	0.33
Library 3	0.10	0.55	0.35	0.05	0.35	0.15	0.60	0.20	0.20	0.40	0.60	0.32	0.60	0.25
Library 4	0.48	0.57	0.61	0.57	0.57	0.70	0.43	0.43	0.26	0.61	0.43	0.51	0.78	0.52
Library 5	0.55	0.36	0.64	0.18	0.64	0.36	0.36	0.45	0.27	0.73	0.55	0.46	0.73	0.00
Library 3	0.25	0.33	0.17	0.08	0.33	0.08	0.58	0.50	0.08	0.33	0.17	0.27	0.08	0.25

Table S13. Split-level performance of individual chemists vs the multi-class classification models for predicting binned reaction yields.



Figure S27. Library-Level Performance of Individual Chemists vs the multi-class classification models for predicting Binned Reaction Yields. Blue represents the chemist while orange represents the model. The naïve baseline (class = 0) is drawn for each library as a horizontal dashed green line.

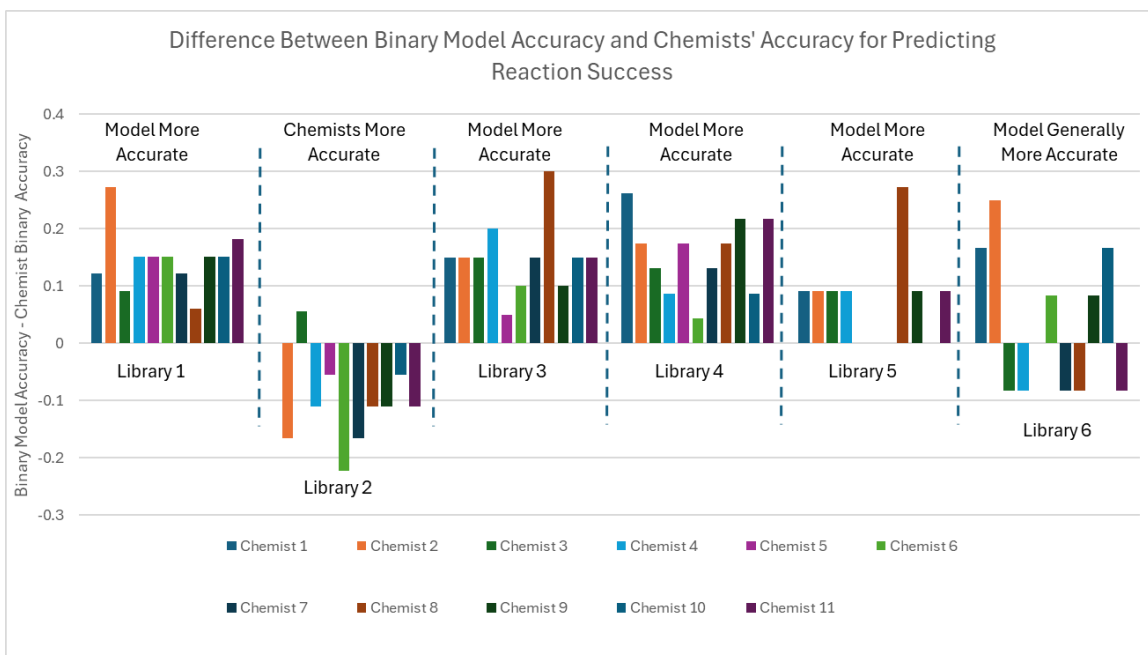


Figure S28. Difference between the binary model's accuracy and the chemists' accuracy for predicting reaction success. Positive values indicate that the model performed better, while negative values indicate that the chemist performed better. Empty values indicate that the chemist performed equal to the model.

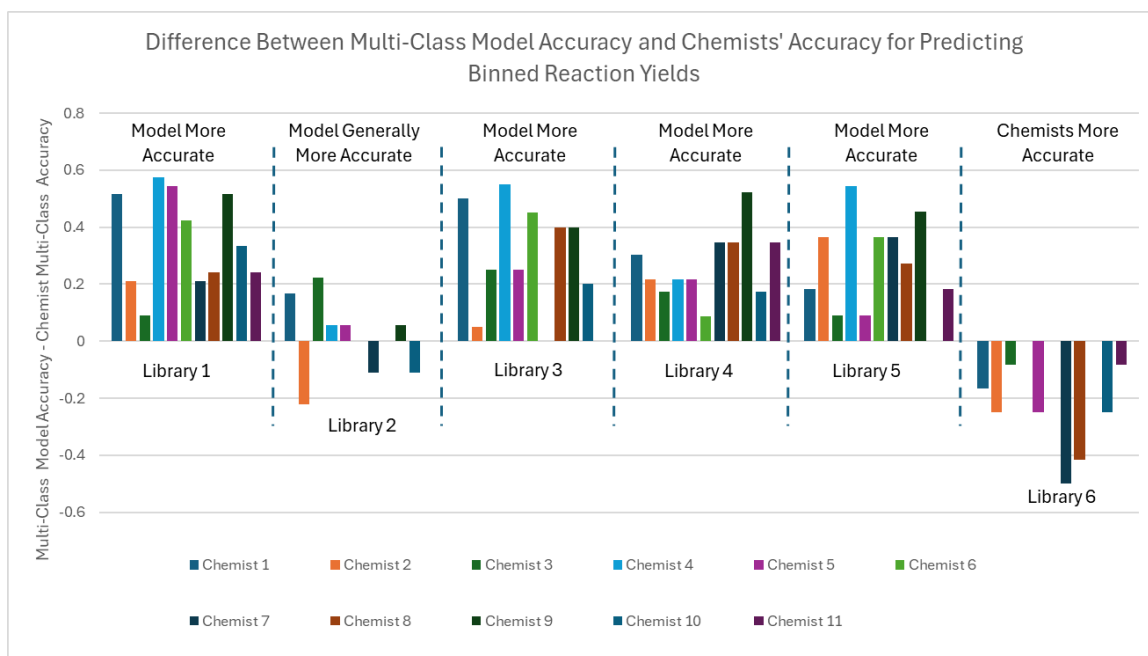


Figure S29. Difference between the multi-class model's accuracy and the chemists' accuracy for predicting binned reaction yields. Positive values indicate that the model performed better, while negative values indicate that the chemist performed better. Empty values indicate that the chemist performed equal to the model.

The “best” overall chemist surveyed was Chemist #7, who achieved the 2nd highest score for both binary and multi-class accuracy of the chemists surveyed with 73.5% accuracy and 49.6% accuracy, respectively. This chemist has >20 years of experience in the field and was among the top-3-scoring chemists 8/12 times (6 libraries x 2 prediction tasks). Chemist #7 achieved greater accuracy than the models in 4/12 instances, in addition to tying with the model in 2/12 instances.

Chemist #3 was the most experienced of those surveyed with >30 years in the field. They generally performed consistently among the others surveyed, appearing 6/12 times in the top-3-scoring chemists, although there were other chemists who achieved higher accuracy for most of the libraries/prediction tasks surveyed. Chemist #3 achieved greater accuracy than the models in 2/12 instances.

Meanwhile, Chemist #5 was the least-experienced medicinal chemist surveyed, with <1 year of experience as a medicinal chemist. Chemist #5 appeared 4/12 times among the top-3-scoring chemists. Overall, they achieved higher accuracy than the models in 2/12 instances and tied with the models in 2/12 instances.

These results highlight the potential for the models to provide consistency in yield prediction during library design, as even the highest-performing chemist was outperformed by the model in 6/12 instances.

5. Experimental Procedures for the Suzuki Cross-Couplings (Prospective Application)

General Procedure 1: Boronic acid/ester monomers (1.5 equiv., 0.15 mmol) were pre-weighed into 4mL vials with magnetic stir bars. A heterogeneous stock solution of the aryl bromide core (0.1 M in 1,4-dioxane) was prepared, along with stock solutions of aqueous Cs₂CO₃ (1M in deionized water) and a partially heterogeneous stock solution of Pd(dppf)Cl₂·CH₂Cl₂ (0.02M). The aryl halide stock solution (0.1M, 1.0 mL, 1.0 equiv., 0.1 mmol) was added to the 4mL vials containing the monomers, followed by the Cs₂CO₃ stock solution (1M, 0.2 mL, 2.0 equiv., 0.2 mmol) and the Pd stock solution (0.02M, 0.5 mL, 10 mol%, 0.01 mmol). The vials were then sealed under normal atmospheric conditions and stirred at 100 °C for 18h. The reactions were then cooled to room temperature and the solvent was removed by blowing nitrogen gas over the vials. The vials were then re-dissolved with 2mL of DMSO/water (9:1) and filtered through a pad of mixed silica/Celite (5%) into a 96-deep well plate via centrifugation at 2,000x g. A small aliquot (10 µL) of each reaction was transferred to a 96-well LCMS plate and diluted further with 140 µL of MeOH and analyzed with UPLC-MS. The reactions with product detected in the MS traces were purified *via* reverse-phase mass-directed preparative HPLC using an ammonium acetate/acetonitrile gradient unless noted otherwise. The identity and purity of the final products were analyzed *via* NMR and UPLC-MS analysis.

General Procedure 2: Boronic acid/ester monomers (1.2 equiv., 0.24 mmol) were pre-weighed into 4mL vials and diluted with 0.7 mL dioxane. A stock solution of the aryl chloride core (0.29 M in 1,4-dioxane) was prepared, along with a stock solution of aqueous Cs₂CO₃ (1M in deionized water). The aryl halide stock solution (0.29M, 0.7 mL, 1.0 equiv., 0.2 mmol) was added to a 0.5-2 mL microwave vial containing a stir bar, meanwhile the Cs₂CO₃ stock solution (1M, 0.6 mL, 3.0 equiv., 0.6 mmol) was added to the monomer dissolved in dioxane. The solution of monomer and base was transferred to the microwave tube *via* pipette and Siliacat DPP-Pd was then added using a calibrated scoop (0.25 mmol/g, 80.0 mg, 10 mol%, 0.02 mmol). The microwave tube was then sealed under normal atmospheric conditions and the reaction was conducted in a microwave reactor under the following conditions: 15 sec pre-stir, absorbance: high, temperature: 150 °C, time: 20 mins (**caution:** at this high temperature, it is possible for the vial to explode if the absorbance

is not set to “high,” as the temperature will otherwise increase too quickly for the instrument to stop applying radiation). The reaction was then cooled to room temperature and the catalyst and any precipitates were filtered off. The solvent was then removed by blowing N₂. The crude residue was re-dissolved with 2mL of DMSO/water (9:1) and filtered once more through a pad of mixed silica/Celite (5%) into a 96-deep well plate via centrifugation at 2,000x g. A small aliquot (10 µL) of each reaction was transferred to a 96-well LCMS plate and diluted further with 140 µL of MeOH and analyzed *via* UPLC-MS. The reactions with product detected in the MS traces were purified *via* reverse-phase mass-directed preparative HPLC using an ammonium acetate/acetonitrile gradient unless noted otherwise. The identity and purity of the final products were assessed *via* NMR and UPLC-MS analysis.

6. Details and Workflow for Prospective Monomer Replacement Application

Steps for calculating pairwise shape and electrostatic similarity values for Suzuki monomers:

- 1) Extracted all monomers from the parallel library dataset. Filtered to organoborane monomers only.
Notes:
 - a. Did not consider (pseudo)halide monomers for this exercise since such libraries were few in number.
 - b. Did not include monomers outside of AbbVie’s retrospective parallel library dataset for this proof-of-concept study.
- 2) Converted all boronic esters to boronic acids, because we wanted the similarity calculations to be based on the R-group only, and not depend on the type of the boron functional group.
- 3) Calculated shape and electrostatic similarity values using the code associated with a previous report.¹⁵ Each monomer pair was aligned along the “B(OH)OH” fragment and the “EmbedAlignConstrainedScore()” function was used with the following settings:
 - a. renormalize=True
 - b. prbNumConfs=50
 - c. refNumConfs=50
 - d. metric='carbo'
 - e. integrate='gauss'
 - f. partialCharges='ml'
 - g. getBestESP=False
- 4) Net shape and electrostatic similarity values were calculated as a product of the respective shape similarity and electrostatic potential similarity values obtained from the code.
- 5) Similarity values were merged with predicted yields for the reaction of the replacement monomers with the core of interest to generate the output files used for monomer replacement.
- 6) Please see the GitHub repository (src/utls/monomer_replacement.py) for the code that was used to generate the similarity scores between 2 molecules.

Using this approach, the query monomers for those predicted to be unsuccessful by the binary RF model (fp + DFT; library (and the core) was unseen by the model) or those that previously were unsuccessful

during the library synthesis (RF, fp + DFT, library was seen by the model) were used to generate files containing the 10 most similar monomers and their predicted yields with the regression models. The 10 most similar monomers for each query monomer were ranked in the order of their predicted yields (using either of the two RF models, depending on the query monomer), and the monomer with the highest predicted yield was selected for replacement. Replacement monomers were not considered if the corresponding product was present in the original library, and duplicated replacement monomers were changed by selecting the next monomer with the highest predicted yield that was not present in the original library.

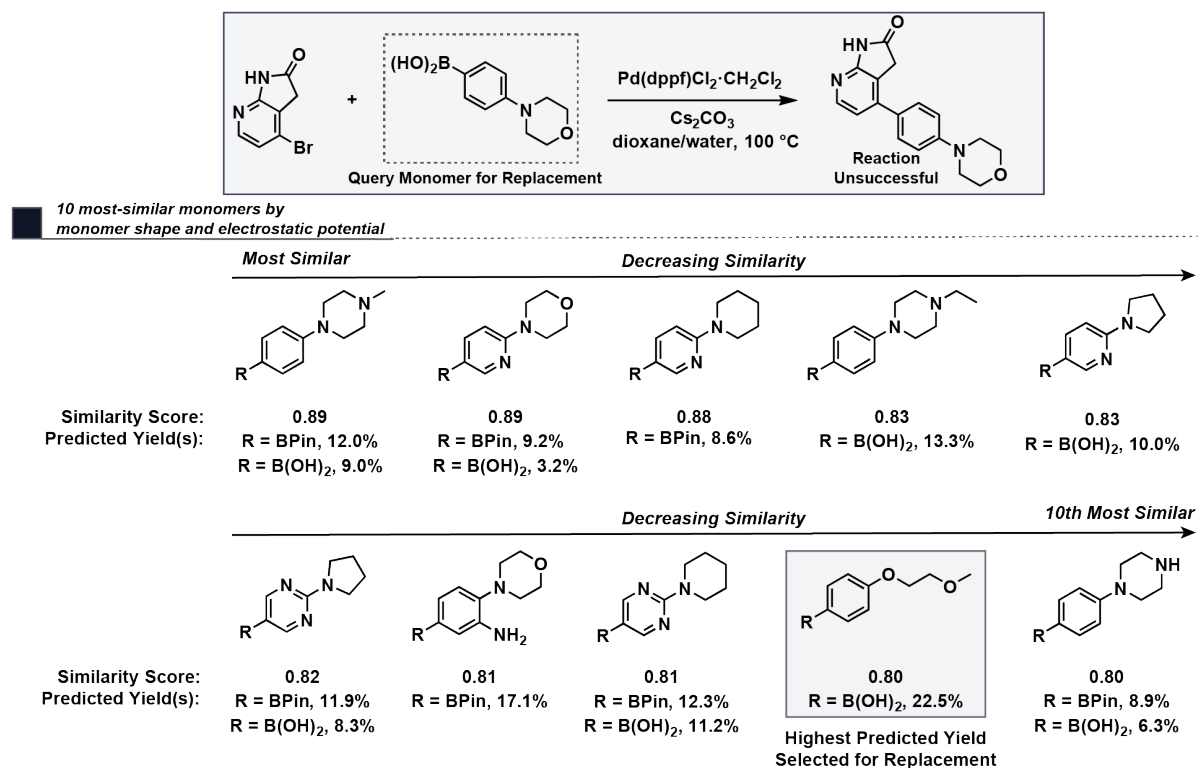


Figure S30. Example of replacement monomer selection for an unsuccessful monomer (*monomer rescue*)

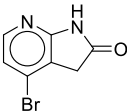
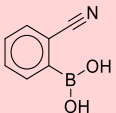
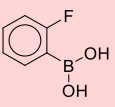
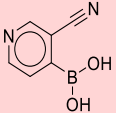
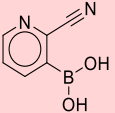
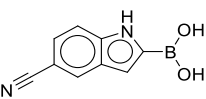
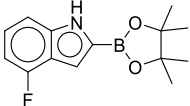
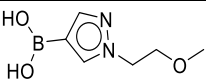
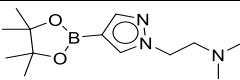
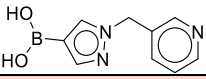
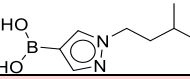
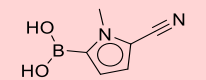
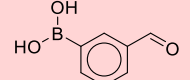
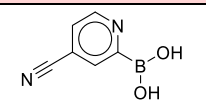
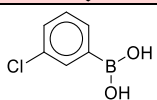
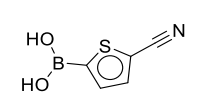
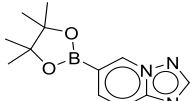
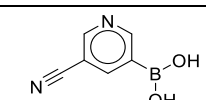
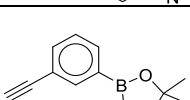
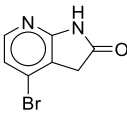
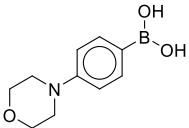
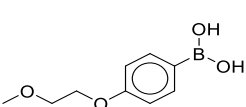
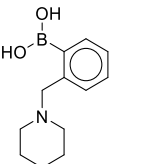
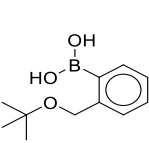
Core Structure	Original Monomer Structure	Replacement Monomer Structure	Similarity Score	Predicted Yield	Isolated Yield
			0.83	27%	0%
			0.91	17%	0%
			0.7	17%	14%
			0.87	18%	9%
			0.77	30%	11%
			0.75	22%	0%
			0.84	19%	40%
			0.77	17%	21%
			0.84	20%	6% ^a
Note: this monomer was successfully used during the original synthesis					

Table S14. Monomer Prioritization replacement monomers, predicted yields, isolated yields, and similarity scores. Similarity calculations were obtained by using the boronic acid structures of all replacement molecules. Unsuccessful replacements are highlighted in red. ^aThis compound was repurified and therefore the final isolated yield is expected to be lower than if it was not.

Core Structure	Original Monomer Structure	Replacement Monomer Structure	Sim. Score	Predicted Yield	Iso. Yield
			0.8	23%	20%
			0.76	28%	21% ^a

		0.66	22%	26%
		0.9	14%	29%
		0.83	22%	0%
		0.7	28%	18%
		0.87	24%	26%
		0.92	15%	32%
		0.72	23%	43%
		0.85	14%	7%
		0.87	23%	28%
		0.82	13%	0%
		0.62	23%	21%
		0.79	30%	32%
		0.7	18%	14%
		0.73	25%	26%
		0.87	18%	9%

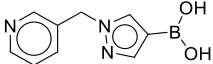
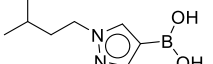
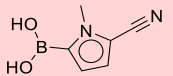
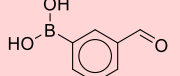
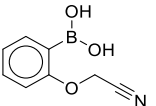
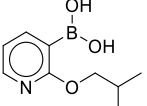
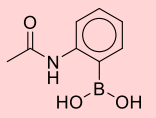
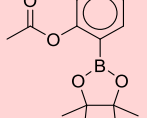
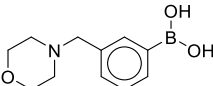
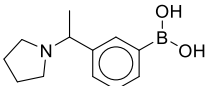
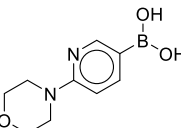
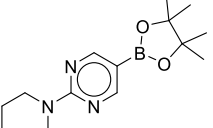
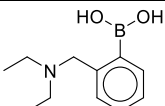
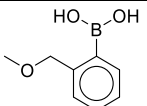
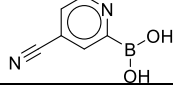
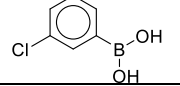
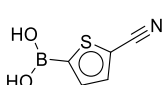
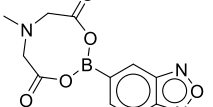
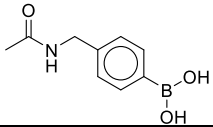
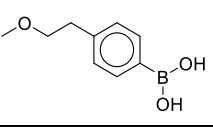
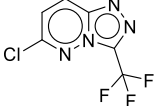
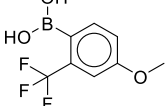
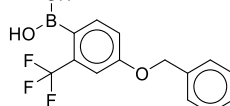
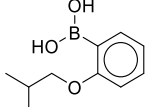
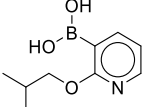
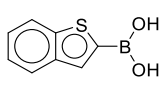
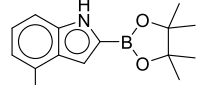
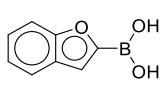
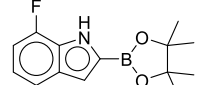
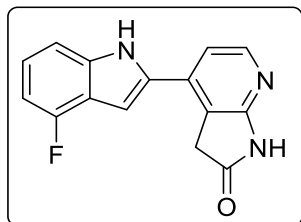
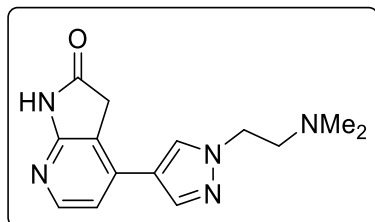
			0.77	19%	11%
			0.75	19%	0%
			0.73	24%	30%
			0.77	17%	0% ^b
			0.78	39%	40%
			0.82	12%	28%
			0.72	26%	41%
			0.84	17%	40%
			0.78	14%	20%
			0.82	30%	31%
			0.70	5%	2%
			0.89	11%	12%
			0.79	17%	21%
			0.78	10%	22%

Table S15. Monomer Rescue replacement monomers, predicted yields, isolated yields, and similarity scores. Similarity calculations were obtained by using the boronic acid structures of all replacement molecules. ^aThis compound was repurified and therefore the final isolated yield is expected to be lower than if it was not. ^bA Suzuki product was formed in this reaction according to the LCMS data; however, the Suzuki product's ester underwent hydrolysis to the phenol during the reaction. Consequently, the compound was not collected during purification as the expected *m/z* was not observed.

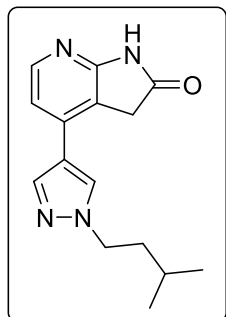
7. Suzuki Product Characterization



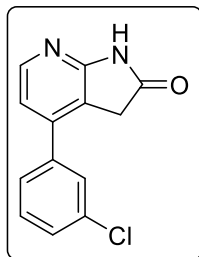
4-(4-fluoro-1H-indol-2-yl)-1,3-dihydro-2H-pyrrolo[2,3-b]pyridin-2-one (1). Synthesized from **Core A** and **Monomer 3** according to general procedure #1 for Suzuki reactions. 0.1 mmol scale: 14% yield (3.7 mg). Gray solid. ¹H NMR (400 MHz, DMSO-*d*₆) δ 11.92 (d, *J* = 2.4 Hz, 1H), 11.10 (s, 1H), 8.17 (d, *J* = 5.6 Hz, 1H), 7.40 (d, *J* = 5.7 Hz, 1H), 7.32 (d, *J* = 8.1 Hz, 1H), 7.17 (td, *J* = 8.0, 5.3 Hz, 1H), 7.02 (d, *J* = 2.1 Hz, 1H), 6.84 (dd, *J* = 10.6, 7.7 Hz, 1H), 3.91 (s, 2H). ¹³C NMR (101 MHz, DMSO-*d*₆) δ 175.6, 159.2, 155.7 (d, *J* = 245.7 Hz), 146.8, 139.6 (d, *J* = 10.9 Hz), 134.3, 133.9, 123.6 (d, *J* = 7.8 Hz), 117.3 (d, *J* = 22.0 Hz), 115.8, 113.9, 108.3 (d, *J* = 3.6 Hz), 104.3 (d, *J* = 18.3 Hz), 99.4, 36.2). ¹⁹F NMR (376 MHz, DMSO-*d*₆) δ -121.44 (ddd, *J* = 10.6, 5.3, 2.3 Hz). LRMS (ESI) Calcd for C₁₅H₁₀FN₃O [M+H]⁺: 268.1; Found: 268.1.



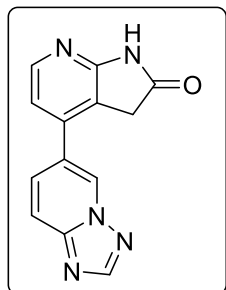
4-(1-(2-(dimethylamino)ethyl)-1H-pyrazol-4-yl)-1,3-dihydro-2H-pyrrolo[2,3-b]pyridin-2-one (2). Synthesized from **Core A** and **Monomer 4** according to general procedure #1 for Suzuki reactions. 0.1 mmol scale: 9% yield (2.3 mg). White solid. ¹H NMR (400 MHz, DMSO-*d*₆) δ 10.95 (s, 1H), 8.32 (d, *J* = 0.8 Hz, 1H), 8.03 – 7.95 (m, 2H), 7.17 (d, *J* = 5.6 Hz, 1H), 4.24 (t, *J* = 6.5 Hz, 2H), 3.71 – 3.67 (m, 2H), 2.68 (t, *J* = 6.5 Hz, 2H), 2.17 (s, 6H). ¹³C NMR (101 MHz, DMSO-*d*₆) δ 175.6, 158.9, 146.3, 137.6, 135.6, 129.9, 117.7, 114.4, 114.3, 58.4, 49.7, 45.1, 36.0. LRMS (ESI) Calcd for C₁₄H₁₇N₅O [M+H]⁺: 272.2; Found: 272.2.



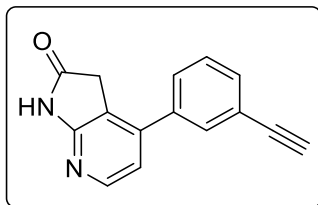
4-(1-isopentyl-1H-pyrazol-4-yl)-1,3-dihydro-2H-pyrrolo[2,3-b]pyridin-2-one (3). Synthesized from **Core A** and **Monomer 5** according to general procedure #1 for Suzuki reactions. 0.1 mmol scale: 11% yield (3.0 mg). Light pink solid. ^1H NMR (400 MHz, DMSO- d_6) δ 10.95 (s, 1H), 8.34 (d, J = 0.9 Hz, 1H), 8.02 – 7.96 (m, 2H), 7.18 (d, J = 5.7 Hz, 1H), 4.17 (dd, J = 7.9, 6.6 Hz, 2H), 3.72 – 3.68 (m, 2H), 1.71 (dt, J = 8.0, 6.7 Hz, 2H), 1.50 (dh, J = 13.4, 6.8 Hz, 1H), 0.91 (d, J = 6.6 Hz, 6H). ^{13}C NMR (101 MHz, DMSO- d_6) δ 175.6, 158.8, 146.3, 137.5, 135.6, 129.5, 117.7, 114.4, 114.3, 49.9, 38.6, 36.0, 25.0, 22.2. LRMS (ESI) Calcd for $\text{C}_{15}\text{H}_{18}\text{N}_4\text{O}$ $[\text{M}+\text{H}]^+$: 271.2; Found: 271.2.



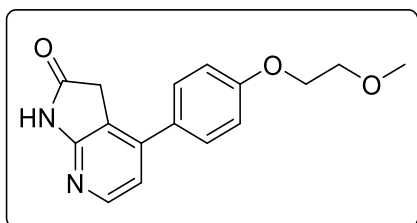
4-(3-chlorophenyl)-1,3-dihydro-2H-pyrrolo[2,3-b]pyridin-2-one (4). Synthesized from **Core A** and **Monomer 7** according to general procedure #1 for Suzuki reactions. 0.1 mmol scale: 40% yield (9.7 mg). Tan solid. ^1H NMR (400 MHz, DMSO- d_6) δ 11.10 (s, 1H), 8.14 (d, J = 5.6 Hz, 1H), 7.72 (q, J = 1.4 Hz, 1H), 7.63 (ddd, J = 6.3, 2.8, 1.7 Hz, 1H), 7.54 – 7.50 (m, 2H), 7.10 (d, J = 5.5 Hz, 1H), 3.76 (s, 2H). ^{13}C NMR (101 MHz, DMSO- d_6) δ 175.5, 158.9, 146.8, 142.2, 139.1, 133.7, 130.8, 128.8, 127.6, 126.7, 117.7, 116.7, 35.2. LRMS (ESI) Calcd for $\text{C}_{13}\text{H}_9\text{ClN}_2\text{O}$ $[\text{M}+\text{H}]^+$: 245.0; Found: 245.1.



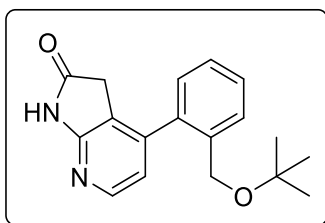
4-([1,2,4]triazolo[1,5-a]pyridin-6-yl)-1,3-dihydro-2H-pyrrolo[2,3-b]pyridin-2-one (5). Synthesized from **Core A** and **Monomer 8** according to general procedure #1 for Suzuki reactions. 0.1 mmol scale: 21% yield (5.3 mg). Off-white solid. ^1H NMR (400 MHz, DMSO- d_6) δ 11.15 (s, 1H), 9.35 (s, 1H), 8.59 (s, 1H), 8.18 (d, J = 4.4 Hz, 1H), 7.97 (s, 2H), 7.23 (d, J = 4.5 Hz, 1H), 3.91 – 3.86 (m, 2H). ^{13}C NMR (101 MHz, DMSO- d_6) δ 175.6, 159.0, 154.8, 149.6, 147.0, 139.5, 130.4, 128.2, 124.3, 118.1, 116.9, 116.4, 35.2. LRMS (ESI) Calcd for $\text{C}_{13}\text{H}_9\text{N}_5\text{O}$ $[\text{M}+\text{H}]^+$: 252.1; Found: 252.1.



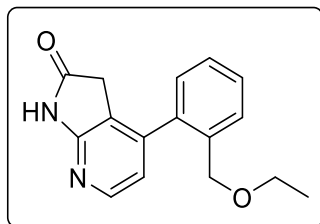
4-(3-ethynylphenyl)-1,3-dihydro-2H-pyrrolo[2,3-b]pyridin-2-one (6). Synthesized from **Core A** and **Monomer 9** according to general procedure #1 for Suzuki reactions. The compound was re-purified using the Agilent system with a formic acid gradient. 0.1 mmol scale: 6% yield (1.3 mg). White solid. ^1H NMR (400 MHz, DMSO- d_6) δ 11.09 (s, 1H), 8.13 (d, J = 5.5 Hz, 1H), 7.77 – 7.67 (m, 2H), 7.60 – 7.48 (m, 2H), 7.10 (d, J = 5.5 Hz, 1H), 4.28 (s, 1H), 3.76 (s, 2H). ^{13}C NMR (101 MHz, DMSO- d_6) δ 175.6, 158.9, 146.8, 142.6, 137.5, 132.0, 130.8, 129.4, 128.5, 122.5, 117.6, 116.7, 83.0, 81.5, 35.3. LRMS (ESI) Calcd for $\text{C}_{15}\text{H}_{10}\text{N}_2\text{O}$ $[\text{M}+\text{H}]^+$: 235.1; Found: 235.1.



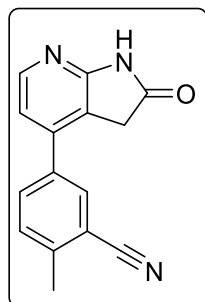
4-(4-(2-methoxyethoxy)phenyl)-1,3-dihydro-2H-pyrrolo[2,3-b]pyridin-2-one (7). Synthesized from **Core A** and **Monomer 10** according to general procedure #1 for Suzuki reactions. 0.1 mmol scale: 20% yield (5.7 mg). Tan solid. ^1H NMR (400 MHz, DMSO- d_6) δ 10.99 (br s, 1H), 8.08 (d, J = 5.5 Hz, 1H), 7.66 – 7.57 (m, 2H), 7.10 – 7.01 (m, 3H), 4.18 – 4.12 (m, 2H), 3.73 (s, 2H), 3.70 – 3.64 (m, 2H), 3.31 (s, 3H). ^{13}C NMR (101 MHz, DMSO- d_6) δ 175.7, 159.1, 158.9, 146.6, 143.3, 129.3, 129.1, 116.7, 116.4, 114.9, 70.4, 67.1, 58.2, 35.6. LRMS (ESI) Calcd for $\text{C}_{16}\text{H}_{16}\text{N}_2\text{O}_3$ $[\text{M}+\text{H}]^+$: 285.1; Found: 285.1.



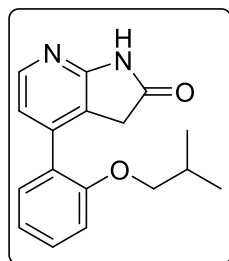
4-(2-(tert-butoxymethyl)phenyl)-1,3-dihydro-2H-pyrrolo[2,3-b]pyridin-2-one (8). Synthesized from **Core A** and **Monomer 11** according to general procedure #1 for Suzuki reactions, except the compound was re-purified using a formic acid/acetonitrile gradient. 0.1 mmol scale: 21% yield (6.3 mg). Pale yellow oil. ^1H NMR (499 MHz, DMSO- d_6) δ 11.05 (s, 1H), 8.11 – 8.05 (m, 1H), 7.53 – 7.46 (m, 1H), 7.39 (dtd, J = 17.2, 7.4, 1.6 Hz, 2H), 7.31 – 7.24 (m, 1H), 6.87 (d, J = 5.4 Hz, 1H), 4.15 (s, 2H), 3.35 (s, 2H; peak overlaps with water at higher concentration), 1.01 (s, 9H). ^{13}C NMR (101 MHz, DMSO- d_6) δ 175.5, 158.3, 145.7, 144.3, 137.2, 136.3, 130.0, 128.3, 128.3, 127.6, 119.2, 118.4, 72.8, 61.5, 34.8, 27.0. LRMS (ESI) Calcd for $\text{C}_{18}\text{H}_{20}\text{N}_2\text{O}_2$ $[\text{M}+\text{H}]^+$: 297.2; Found: 297.2.



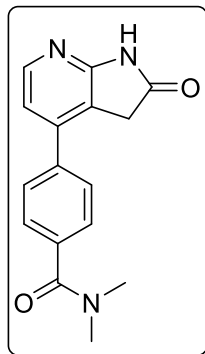
4-(2-(ethoxymethyl)phenyl)-1,3-dihydro-2H-pyrrolo[2,3-b]pyridin-2-one (9). Synthesized from **Core A** and **Monomer 12** according to general procedure #1 for Suzuki reactions. 0.1 mmol scale: 26% yield (7.0 mg). Oily brown solid. ^1H NMR (400 MHz, $\text{DMSO}-d_6$, 90 °C) δ 10.73 (br s, 1H), 8.08 (dt, J = 5.4, 0.9 Hz, 1H), 7.55 – 7.49 (m, 1H), 7.40 (dtd, J = 17.2, 7.4, 1.6 Hz, 2H), 7.29 – 7.24 (m, 1H), 6.83 (d, J = 5.5 Hz, 1H), 4.27 (s, 2H), 3.36 (q, J = 7.0 Hz, 2H), 3.31 (s, 2H), 1.04 (t, J = 6.9 Hz, 3H). ^{13}C NMR (101 MHz, $\text{DMSO}-d_6$) δ 175.5, 158.5, 145.8, 144.0, 136.7, 135.6, 129.2, 128.4, 128.4, 127.7, 119.0, 118.2, 69.4, 65.1, 34.7, 14.9. LRMS (ESI) Calcd for $\text{C}_{16}\text{H}_{16}\text{N}_2\text{O}_2$ $[\text{M}+\text{H}]^+$: 269.1; Found: 269.2.



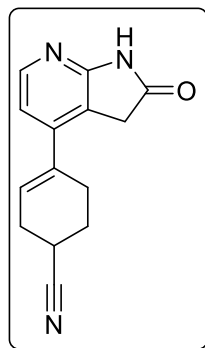
4-(2-(ethoxymethyl)phenyl)-1,3-dihydro-2H-pyrrolo[2,3-b]pyridin-2-one (10). Synthesized from **Core A** and **Monomer 13** according to general procedure #1 for Suzuki reactions. 0.1 mmol scale: 29% yield (7.1 mg). Gray solid. ^1H NMR (600 MHz, $\text{DMSO}-d_6$) δ 11.11 (s, 1H), 8.14 (dd, J = 5.5, 0.9 Hz, 1H), 8.08 (d, J = 2.0 Hz, 1H), 7.89 (dd, J = 8.0, 2.1 Hz, 1H), 7.59 (dt, J = 8.1, 0.7 Hz, 1H), 7.12 (d, J = 5.6 Hz, 1H), 3.79 (d, J = 0.9 Hz, 2H), 2.53 (s, 3H). ^{13}C NMR (151 MHz, $\text{DMSO}-d_6$) δ 175.6, 159.0, 146.9, 142.2, 141.6, 135.3, 132.5, 131.6, 131.1, 117.64, 117.59, 116.6, 112.6, 35.2, 19.8. LRMS (ESI) Calcd for $\text{C}_{15}\text{H}_{11}\text{N}_3\text{O}$ $[\text{M}+\text{H}]^+$: 250.1; Found: 250.1.



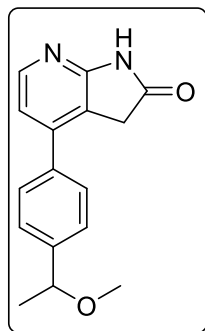
4-(2-isobutoxyphenyl)-1,3-dihydro-2H-pyrrolo[2,3-b]pyridin-2-one (11). Synthesized from **Core A** and **Monomer 14** according to general procedure #1 for Suzuki reactions. 0.1 mmol scale: 18% yield (5.2 mg). Oily brown solid. ^1H NMR (400 MHz, $\text{DMSO}-d_6$, 90 °C) δ 8.06 (d, J = 5.4 Hz, 1H), 7.44 – 7.36 (m, 1H), 7.31 (dd, J = 7.5, 1.8 Hz, 1H), 7.13 (d, J = 8.1 Hz, 1H), 7.05 (t, J = 7.5 Hz, 1H), 6.90 (d, J = 5.5 Hz, 1H), 3.78 (d, J = 6.4 Hz, 2H), 3.40 (s, 2H), 1.89 (dp, J = 13.3, 6.6 Hz, 1H), 0.84 (d, J = 6.7 Hz, 6H). ^{13}C NMR (151 MHz, $\text{DMSO}-d_6$) δ 175.5, 158.2, 155.4, 145.6, 142.1, 130.2, 129.9, 126.0, 120.6, 119.0 (2 peaks overlapping), 112.9, 74.2, 35.3, 27.8, 19.0. LRMS (ESI) Calcd for $\text{C}_{17}\text{H}_{18}\text{N}_2\text{O}_2$ $[\text{M}+\text{H}]^+$: 283.1; Found: 283.2.



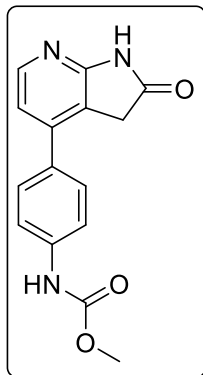
N,N-dimethyl-4-(2-oxo-2,3-dihydro-1H-pyrrolo[2,3-b]pyridin-4-yl)benzamide (12). Synthesized from **Core A** and **Monomer 15** according to general procedure #1 for Suzuki reactions. 0.1 mmol scale: 26% yield (7.4 mg). Oily gray solid. $^1\text{H NMR}$ (400 MHz, $\text{DMSO-}d_6$) δ 11.09 (s, 1H), 8.14 (d, $J = 5.5$ Hz, 1H), 7.77 – 7.69 (m, 2H), 7.56 – 7.48 (m, 2H), 7.12 (d, $J = 5.6$ Hz, 1H), 3.78 (s, 2H), 3.00 (s, 3H), 2.95 (s, 3H). $^{13}\text{C NMR}$ (101 MHz, $\text{DMSO-}d_6$) δ 175.6, 169.6, 159.0, 146.8, 142.9, 137.7, 136.8, 127.8, 127.5, 117.5, 116.6, 35.4, 34.8. **LRMS** (ESI) Calcd for $\text{C}_{16}\text{H}_{15}\text{N}_3\text{O}_2$ $[\text{M}+\text{H}]^+$: 282.1; Found: 282.2.



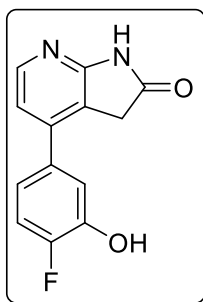
4-(2-oxo-2,3-dihydro-1H-pyrrolo[2,3-b]pyridin-4-yl)cyclohex-3-ene-1-carbonitrile (13). Synthesized from **Core A** and **Monomer 16** according to general procedure #1 for Suzuki reactions. 0.1 mmol scale: 32% yield (7.6 mg). Off-white solid. $^1\text{H NMR}$ (400 MHz, $\text{DMSO-}d_6$) δ 10.96 (s, 1H), 8.00 (d, $J = 5.6$ Hz, 1H), 6.87 (d, $J = 5.6$ Hz, 1H), 6.14 (td, $J = 3.8, 1.9$ Hz, 1H), 3.61 (s, 2H), 3.11 (dtd, $J = 12.2, 6.2, 5.8, 3.4$ Hz, 1H), 2.59 (ddt, $J = 18.7, 6.4, 2.9$ Hz, 1H), 2.43 (dp, $J = 8.4, 2.7, 2.2$ Hz, 3H), 2.02 (dtd, $J = 12.4, 6.0, 3.1$ Hz, 1H), 1.92 (ddt, $J = 12.9, 8.5, 6.4$ Hz, 1H). $^{13}\text{C NMR}$ (101 MHz, $\text{DMSO-}d_6$) δ 175.5, 158.7, 146.4, 144.4, 133.9, 126.1, 122.6, 116.0, 114.9, 35.9, 28.2, 24.8, 24.7, 23.2. **LRMS** (ESI) Calcd for $\text{C}_{14}\text{H}_{13}\text{N}_3\text{O}$ $[\text{M}+\text{H}]^+$: 240.1; Found: 240.1.



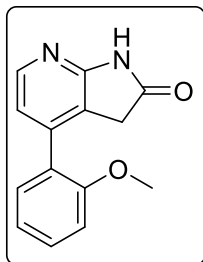
4-(4-(1-methoxyethyl)phenyl)-1,3-dihydro-2H-pyrrolo[2,3-b]pyridin-2-one (14). Synthesized from **Core A** and **Monomer 17** according to general procedure #1 for Suzuki reactions. 0.1 mmol scale: 43% yield (11.6 mg). Light purple solid. ^1H NMR (600 MHz, $\text{DMSO-}d_6$) δ 11.06 (s, 1H), 8.11 (dt, J = 5.5, 0.8 Hz, 1H), 7.69 – 7.64 (m, 2H), 7.46 – 7.40 (m, 2H), 7.08 (dd, J = 5.6, 0.8 Hz, 1H), 4.37 (q, J = 6.4 Hz, 1H), 3.76 (d, J = 0.9 Hz, 2H), 3.15 (d, J = 0.6 Hz, 3H), 1.35 (dd, J = 6.4, 0.6 Hz, 3H). ^{13}C NMR (151 MHz, $\text{DMSO-}d_6$) δ 175.6, 158.9, 146.7, 144.4, 143.4, 135.9, 127.9, 126.6, 117.2, 116.7, 78.2, 55.9, 35.5, 23.4. LRMS (ESI) Calcd for $\text{C}_{16}\text{H}_{16}\text{N}_2\text{O}_2$ $[\text{M}+\text{H}]^+$: 269.1; Found: 269.2.



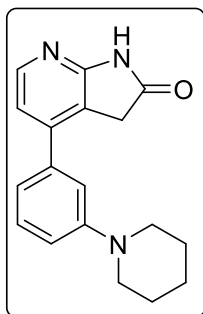
methyl (4-(2-oxo-2,3-dihydro-1H-pyrrolo[2,3-b]pyridin-4-yl)phenyl)carbamate (15). Synthesized from **Core A** and **Monomer 18** according to general procedure #1 for Suzuki reactions. 0.1 mmol scale: 7% yield (1.9 mg). Purple-tan solid. ^1H NMR (400 MHz, $\text{DMSO-}d_6$) δ 11.02 (s, 1H), 9.85 (s, 1H), 8.09 (d, J = 5.6 Hz, 1H), 7.63 (d, J = 8.9 Hz, 2H), 7.58 (d, J = 8.9 Hz, 2H), 7.07 (d, J = 5.5 Hz, 1H), 3.75 (s, 2H), 3.69 (s, 3H). ^{13}C NMR (101 MHz, $\text{DMSO-}d_6$) δ 175.7, 158.9, 154.0, 146.7, 143.2, 139.9, 130.6, 128.5, 118.3, 116.7, 116.4, 51.8, 35.6. LRMS (ESI) Calcd for $\text{C}_{15}\text{H}_{13}\text{N}_3\text{O}_3$ $[\text{M}+\text{H}]^+$: 284.1; Found: 284.1.



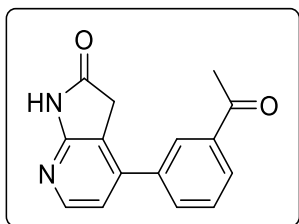
4-(4-fluoro-3-hydroxyphenyl)-1,3-dihydro-2H-pyrrolo[2,3-b]pyridin-2-one (16). Synthesized from **Core A** and **Monomer 19** according to general procedure #1 for Suzuki reactions. 0.1 mmol scale: 28% yield (6.8 mg). Tan solid. ^1H NMR (400 MHz, $\text{DMSO-}d_6$) δ 11.05 (s, 1H), 10.13 (br s, 1H), 8.09 (d, J = 5.5 Hz, 1H), 7.28 – 7.18 (m, 2H), 7.09 (ddd, J = 8.4, 4.3, 2.3 Hz, 1H), 7.01 (d, J = 5.6 Hz, 1H), 3.69 (s, 2H). ^{13}C NMR (101 MHz, $\text{DMSO-}d_6$) δ 175.6, 158.9, 151.4 (d, J = 243.7 Hz), 146.7, 145.2 (d, J = 12.4 Hz), 142.9, 133.5 (d, J = 3.6 Hz), 119.1 (d, J = 6.8 Hz), 117.4 (d, J = 3.5 Hz), 117.1, 116.6 (d, J = 18.4 Hz), 116.5, 35.6. ^{19}F NMR (376 MHz, $\text{DMSO-}d_6$) δ -135.29 (ddd, J = 11.5, 8.5, 4.2 Hz). LRMS (ESI) Calcd for $\text{C}_{13}\text{H}_9\text{FN}_2\text{O}_2$ $[\text{M}+\text{H}]^+$: 245.1; Found: 245.1.



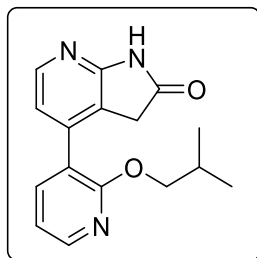
4-(2-methoxyphenyl)-1,3-dihydro-2H-pyrrolo[2,3-b]pyridin-2-one (17). Synthesized from **Core A** and **Monomer 21** according to general procedure #1 for Suzuki reactions. 0.1 mmol scale: 21% yield (5.1 mg). Oily brown solid. $^1\text{H NMR}$ (400 MHz, $\text{DMSO}-d_6$) δ 11.00 (s, 1H), 8.07 (d, $J = 5.5$ Hz, 1H), 7.43 (ddd, $J = 8.3, 7.3, 1.7$ Hz, 1H), 7.33 (dd, $J = 7.5, 1.7$ Hz, 1H), 7.15 (dd, $J = 8.4, 1.0$ Hz, 1H), 7.05 (td, $J = 7.4, 1.0$ Hz, 1H), 6.90 (d, $J = 5.5$ Hz, 1H), 3.79 (s, 3H), 3.41 (d, $J = 0.9$ Hz, 2H). $^{13}\text{C NMR}$ (101 MHz, $\text{DMSO}-d_6$) δ 175.6, 158.2, 155.9, 145.9, 142.1, 130.3, 130.1, 125.7, 120.7, 119.0, 118.9, 111.7, 55.4, 35.4. **LRMS** (ESI) Calcd for $\text{C}_{14}\text{H}_{12}\text{N}_2\text{O}_2$ $[\text{M}+\text{H}]^+$: 241.1; Found: 241.1.



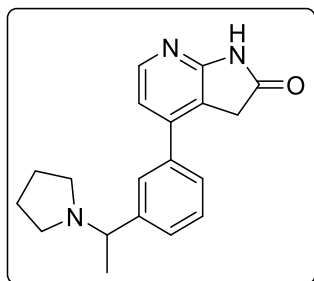
4-(3-(piperidin-1-yl)phenyl)-1,3-dihydro-2H-pyrrolo[2,3-b]pyridin-2-one (18). Synthesized from **Core A** and **Monomer 22** according to general procedure #1 for Suzuki reactions. 0.1 mmol scale: 32% yield (9.4 mg). Off-white solid. $^1\text{H NMR}$ (400 MHz, $\text{DMSO}-d_6$) δ 11.03 (s, 1H), 8.10 (d, $J = 5.5$ Hz, 1H), 7.30 (t, $J = 7.9$ Hz, 1H), 7.10 (t, $J = 2.1$ Hz, 1H), 7.06 (d, $J = 5.5$ Hz, 1H), 6.99 (dd, $J = 7.9, 2.1$ Hz, 2H), 3.72 (s, 2H), 3.23 – 3.16 (m, 4H), 1.67 – 1.58 (m, 4H), 1.54 (qd, $J = 5.7, 5.2, 1.9$ Hz, 2H). $^{13}\text{C NMR}$ (101 MHz, $\text{DMSO}-d_6$) δ 175.7, 158.8, 151.9, 146.6, 144.5, 137.7, 129.5, 117.7, 117.2, 116.9, 116.1, 114.9, 49.4, 35.5, 25.2, 23.9. **LRMS** (ESI) Calcd for $\text{C}_{18}\text{H}_{19}\text{N}_3\text{O}$ $[\text{M}+\text{H}]^+$: 294.2; Found: 294.0.



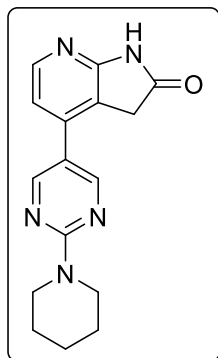
4-(3-acetylphenyl)-1,3-dihydro-2H-pyrrolo[2,3-b]pyridin-2-one (19). Synthesized from **Core A** and **Monomer 23** according to general procedure #1 for Suzuki reactions. 0.1 mmol scale: 26% yield (6.6 mg). Brown solid. $^1\text{H NMR}$ (400 MHz, $\text{DMSO}-d_6$) δ 11.11 (s, 1H), 8.16 (dd, $J = 3.8, 1.9$ Hz, 2H), 8.01 (dt, $J = 7.8, 1.4$ Hz, 1H), 7.96 – 7.90 (m, 1H), 7.65 (t, $J = 7.8$ Hz, 1H), 7.14 (d, $J = 5.5$ Hz, 1H), 3.76 (s, 2H), 2.65 (s, 3H). $^{13}\text{C NMR}$ (101 MHz, $\text{DMSO}-d_6$) δ 197.9, 175.6, 159.0, 146.9, 142.8, 137.5, 137.4, 132.4, 129.4, 128.3, 127.6, 117.7, 116.8, 35.3, 26.9. **LRMS** (ESI) Calcd for $\text{C}_{15}\text{H}_{12}\text{N}_2\text{O}_2$ $[\text{M}+\text{H}]^+$: 253.1; Found: 253.1.



4-(2-isobutoxypyridin-3-yl)-1,3-dihydro-2H-pyrrolo[2,3-b]pyridin-2-one (20). Synthesized from **Core A** and **Monomer 24** according to general procedure #1 for Suzuki reactions. 0.1 mmol scale: 30% yield (8.5 mg). Light brown solid. ^1H NMR (600 MHz, DMSO- d_6) δ 11.06 (s, 1H), 8.22 (dd, J = 5.0, 1.9 Hz, 1H), 8.12 – 8.08 (m, 1H), 7.83 (dd, J = 7.3, 1.9 Hz, 1H), 7.10 (dd, J = 7.3, 5.0 Hz, 1H), 6.99 (d, J = 5.5 Hz, 1H), 4.09 (d, J = 6.6 Hz, 2H), 3.51 (d, J = 0.9 Hz, 2H), 1.95 (hept, J = 6.7 Hz, 1H), 0.87 (d, J = 6.7 Hz, 6H). ^{13}C NMR (151 MHz, DMSO- d_6) δ 175.5, 159.6, 158.4, 147.4, 145.9, 140.2, 139.2, 119.8, 118.9, 118.5, 117.2, 71.9, 35.1, 27.5, 19.1. LRMS (ESI) Calcd for $\text{C}_{16}\text{H}_{17}\text{N}_3\text{O}_2$ $[\text{M}+\text{H}]^+$: 284.1; Found: 284.2.

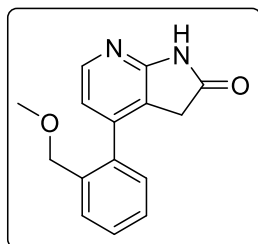


4-(3-(1-(pyrrolidin-1-yl)ethyl)phenyl)-1,3-dihydro-2H-pyrrolo[2,3-b]pyridin-2-one (21). Synthesized from **Core A** and **Monomer 26** according to general procedure #1 for Suzuki reactions. 0.1 mmol scale: 40% yield (19.9 mg) as a tri-acetate salt (4 fractions were isolated with varying amounts of excipient acetic acid salts; the combined average ratio was 3.14:1 AcOH:product). Brown oil. ^1H NMR (400 MHz, DMSO- d_6) δ 8.13 (d, J = 5.5 Hz, 1H), 7.60 (t, J = 1.7 Hz, 1H), 7.53 (dt, J = 7.4, 1.6 Hz, 1H), 7.48 – 7.42 (m, 1H), 7.40 (dt, J = 7.6, 1.5 Hz, 1H), 7.09 (d, J = 5.6 Hz, 1H), 3.76 – 3.72 (m, 2H), 3.34 (q, J = 6.5 Hz, 1H), 2.55 (s, 1H), 2.50 (s, 2H), 2.39 – 2.30 (m, 2H), 1.86 (s, 9H; triacetate salt signal), 1.69 (h, J = 3.1, 2.3 Hz, 4H), 1.35 (d, J = 6.6 Hz, 3H). ^{13}C NMR (101 MHz, DMSO- d_6) δ 175.7, 159.0, 146.8, 146.3, 143.8, 136.9, 129.0, 127.6, 126.5, 126.4, 117.3, 116.8, 64.5, 52.1, 35.5, 23.1, 21.9. LRMS (ESI) Calcd for $\text{C}_{19}\text{H}_{21}\text{N}_3\text{O}$ $[\text{M}+\text{H}]^+$: 308.2; Found: 308.2.

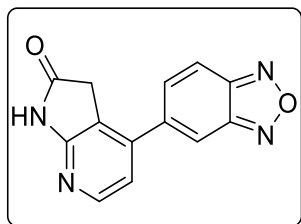


4-(2-(piperidin-1-yl)pyrimidin-5-yl)-1,3-dihydro-2H-pyrrolo[2,3-b]pyridin-2-one (22). Synthesized from **Core A** and **Monomer 27** according to general procedure #1 for Suzuki reactions. 0.1 mmol scale:

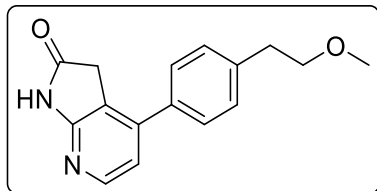
28% yield (8.4 mg). Off-white flakey solid. ^1H NMR (499 MHz, $\text{DMSO-}d_6$) δ 11.03 (s, 1H), 8.69 (s, 2H), 8.08 (d, J = 5.6 Hz, 1H), 7.12 (d, J = 5.6 Hz, 1H), 3.81 (q, J = 3.8, 2.3 Hz, 6H), 1.66 (qd, J = 6.3, 5.8, 2.3 Hz, 2H), 1.54 (td, J = 6.6, 5.9, 3.8 Hz, 4H). ^{13}C NMR (101 MHz, $\text{DMSO-}d_6$) δ 175.6, 160.5, 158.9, 157.0, 146.7, 138.7, 118.0, 116.1, 115.2, 44.3, 35.5, 25.3, 24.2. LRMS (ESI) Calcd for $\text{C}_{16}\text{H}_{17}\text{N}_5\text{O}$ $[\text{M}+\text{H}]^+$: 296.2; Found: 296.2.



4-(2-(methoxymethyl)phenyl)-1,3-dihydro-2H-pyrrolo[2,3-b]pyridin-2-one (23). Synthesized from **Core A** and **Monomer 28** according to general procedure #1 for Suzuki reactions. 0.1 mmol scale: 41% yield (10.3 mg). Tan oily solid. ^1H NMR (400 MHz, $\text{DMSO-}d_6$) δ 11.08 (s, 1H), 8.11 – 8.05 (m, 1H), 7.55 – 7.48 (m, 1H), 7.41 (dtd, J = 17.8, 7.3, 1.6 Hz, 2H), 7.34 – 7.29 (m, 1H), 6.85 (d, J = 5.4 Hz, 1H), 4.22 (s, 2H), 3.36 (s, 2H), 3.18 (s, 3H). ^{13}C NMR (101 MHz, $\text{DMSO-}d_6$) δ 175.5, 158.6, 145.8, 143.9, 136.7, 135.3, 129.2, 128.5, 128.4, 127.8, 118.9, 118.3, 71.3, 57.6, 34.67. LRMS (ESI) Calcd for $\text{C}_{15}\text{H}_{14}\text{N}_2\text{O}_2$ $[\text{M}+\text{H}]^+$: 255.1; Found: 255.1.

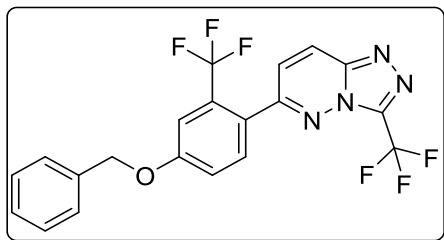


4-(benzo[c][1,2,5]oxadiazol-5-yl)-1,3-dihydro-2H-pyrrolo[2,3-b]pyridin-2-one (24). Synthesized from **Core A** and **Monomer 29** according to general procedure #1 for Suzuki reactions. 0.1 mmol scale: 20% yield (5.0 mg). Light pink solid. ^1H NMR (400 MHz, $\text{DMSO-}d_6$) δ 11.19 (s, 1H), 8.38 (t, J = 1.3 Hz, 1H), 8.24 – 8.13 (m, 2H), 7.88 (dd, J = 9.4, 1.5 Hz, 1H), 7.21 (d, J = 5.5 Hz, 1H), 3.86 (d, J = 0.9 Hz, 2H). ^{13}C NMR (101 MHz, $\text{DMSO-}d_6$) δ 175.5, 159.1, 149.1, 148.4, 147.0, 141.6, 140.9, 133.2, 118.3, 116.9, 116.7, 115.3, 35.2. LRMS (ESI) Calcd for $\text{C}_{13}\text{H}_8\text{N}_4\text{O}_2$ $[\text{M}+\text{H}]^+$: 253.1; Found: 253.1.

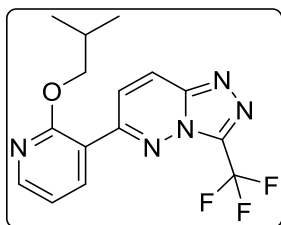


4-(4-(2-methoxyethyl)phenyl)-1,3-dihydro-2H-pyrrolo[2,3-b]pyridin-2-one (25). Synthesized from **Core A** and **Monomer 30** according to general procedure #1 for Suzuki reactions. 0.1 mmol scale: 31% yield (8.2 mg). Gray solid. ^1H NMR (400 MHz, $\text{DMSO-}d_6$) δ 11.04 (s, 1H), 8.11 (d, J = 5.6 Hz, 1H), 7.62 – 7.55 (m, 2H), 7.40 – 7.33 (m, 2H), 7.07 (d, J = 5.6 Hz, 1H), 3.74 (s, 2H), 3.58 (t, J = 6.7 Hz, 2H), 3.25 (s, 3H), 2.86 (t, J = 6.7 Hz, 2H). ^{13}C NMR (101 MHz, $\text{DMSO-}d_6$) δ 175.7, 158.9, 146.7, 143.5, 140.2, 134.6,

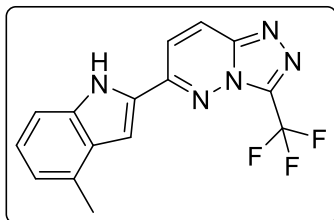
129.4, 127.7, 117.1, 116.7, 72.5, 57.8, 35.5, 35.0. **LRMS** (ESI) Calcd for C₁₆H₁₆N₂O₂ [M+H]⁺: 269.1; Found: 269.2.



6-(4-(benzyloxy)-2-(trifluoromethyl)phenyl)-3-(trifluoromethyl)-[1,2,4]triazolo[4,3-b]pyridazine (26). Synthesized from **Core B** and **Monomer 31** according to general procedure #2 for Suzuki reactions. 0.2 mmol scale: 2% yield (1.4 mg). Colorless oily solid. ¹H NMR (500 MHz, DMSO-*d*₆) δ 8.71 (d, J = 9.7 Hz, 1H), 7.80 (d, J = 9.7 Hz, 1H), 7.73 (d, J = 8.5 Hz, 1H), 7.58 – 7.48 (m, 4H), 7.46 – 7.40 (m, 2H), 7.40 – 7.34 (m, 1H), 5.32 (s, 2H). ¹³C NMR (151 MHz, DMSO-*d*₆) δ 159.7, 155.7, 145.2, 138.3 (d, J = 40.8 Hz), 136.1, 133.7, 128.8 (d, J = 31.0 Hz), 128.6, 128.1, 127.8, 125.3, 125.2 (d, J = 1.9 Hz), 125.1, 123.3 (d, J = 274.1 Hz), 118.6, 118.3 (d, J = 269.8 Hz), 113.9 (q, J = 5.3 Hz), 70.0. ¹⁹F NMR (376 MHz, DMSO-*d*₆) δ -55.65, -62.71. **LRMS** (ESI) Calcd for C₂₀H₁₂F₆N₄O [M+H]⁺: 439.1; Found: 439.2.

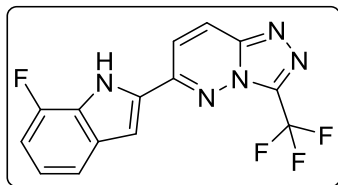


4-(4-(2-hydroxyethyl)phenyl)-1,3-dihydro-2H-pyrrolo[2,3-b]pyridin-2-one (27). Synthesized from **Core B** and **Monomer 24** according to general procedure #2 for Suzuki reactions. 0.2 mmol scale: 12% yield (8.1 mg). Colorless oily solid. ¹H NMR (400 MHz, DMSO-*d*₆) δ 8.66 (d, J = 9.8 Hz, 1H), 8.38 (dd, J = 5.0, 1.9 Hz, 1H), 8.06 (dd, J = 7.5, 2.0 Hz, 1H), 7.99 (d, J = 9.8 Hz, 1H), 7.26 – 7.19 (m, 1H), 4.18 (d, J = 6.5 Hz, 2H), 2.03 (dt, J = 13.3, 6.7 Hz, 1H), 0.94 (d, J = 6.6 Hz, 6H). ¹³C NMR (101 MHz, DMSO-*d*₆) δ 160.6, 154.0, 149.7, 145.3, 140.0, 138.3 (d, J = 40.6 Hz), 125.1, 124.2, 118.5 (d, J = 269.8 Hz), 117.4, 117.1, 72.4, 27.4, 19.1. ¹⁹F NMR (376 MHz, DMSO-*d*₆) δ -62.5. **LRMS** (ESI) Calcd for C₁₅H₁₄F₃N₅O [M+H]⁺: 338.1; Found: 338.2.



4-(4-(2-hydroxyethyl)phenyl)-1,3-dihydro-2H-pyrrolo[2,3-b]pyridin-2-one (28). Synthesized from **Core B** and **Monomer 32** according to general procedure #2 for Suzuki reactions. 0.2 mmol scale: 21% yield (13.4 mg). Yellow solid. ¹H NMR (400 MHz, DMSO-*d*₆) δ 11.69 (s, 1H), 8.55 (ddd, J = 9.9, 2.5, 1.1 Hz, 1H), 8.23 (ddd, J = 9.9, 2.9, 1.4 Hz, 1H), 7.65 (d, J = 2.8 Hz, 1H), 7.40 (d, J = 8.2 Hz, 1H), 7.19 – 7.10 (m, 1H), 6.86 (dd, J = 7.1, 1.8 Hz, 1H), 2.53 (s, 3H). ¹³C NMR (101 MHz, DMSO-*d*₆) δ 148.9, 145.7, 138.4, 138.2 (d, J = 40.1 Hz), 130.7, 130.5, 128.0, 124.9, 124.7, 122.3, 120.1, 118.5 (q, J = 269.7 Hz),

110.2, 106.7, 18.4. ^{19}F NMR (376 MHz, $\text{DMSO}-d_6$) δ -62.0. LRMS (ESI) Calcd for $\text{C}_{15}\text{H}_{10}\text{F}_3\text{N}_5\text{O}_2$ $[\text{M}+\text{H}]^+$: 318.1; Found: 318.2.



6-(7-fluoro-1H-indol-2-yl)-3-(trifluoromethyl)-[1,2,4]triazolo[4,3-b]pyridazine (29). Synthesized from **Core B** and **Monomer 33** according to general procedure #2 for Suzuki reactions. 0.2 mmol scale: 22% yield (13.9 mg). Off-white fluffy solid. ^1H NMR (400 MHz, $\text{DMSO}-d_6$) δ 12.33 (s, 1H), 8.61 (d, J = 9.8 Hz, 1H), 8.23 (d, J = 9.9 Hz, 1H), 7.58 (d, J = 3.0 Hz, 1H), 7.49 (d, J = 7.6 Hz, 1H), 7.06 (ddt, J = 12.5, 7.6, 6.2 Hz, 2H). ^{13}C NMR (101 MHz, $\text{DMSO}-d_6$) δ 150.4, 148.7, 146.8 (d, J = 243.3 Hz), 138.3 (d, J = 40.4 Hz), 132.7 (d, J = 1.1 Hz), 131.4 (d, J = 5.6 Hz), 126.5 (d, J = 13.4 Hz), 125.1, 122.1, 120.6 (d, J = 6.1 Hz), 118.46 (q, J = 269.8 Hz), 117.65 (d, J = 3.7 Hz), 109.06 (d, J = 16.2 Hz), 107.58 (d, J = 2.3 Hz). ^{19}F NMR (376 MHz, $\text{DMSO}-d_6$) δ -62.0 (d, J = 2.3 Hz), -130.0 (ddd, J = 11.2, 5.1, 2.9 Hz). LRMS (ESI) Calcd for $\text{C}_{14}\text{H}_7\text{F}_4\text{N}_5$ $[\text{M}+\text{H}]^+$: 322.1; Found: 322.2.

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9. ^1H , ^{13}C , and ^{19}F NMR Spectra

